An Internship Report on

Development of a Dynamic Ride-Sharing and Carpooling Platform

&

Infosys Springboard.

BY

Rajsingh Ganeshsingh Thakur

Under the Guidance

of

Ms. Savita S. Wagre

(Department of Computer Science And Engineering)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Mahatma Gandhi Mission's College of Engineering, Nanded (M.S.)

Academic Year 2025-26

An Internship Report on

"Development of a Dynamic Ride-Sharing and Carpooling Platform"
&
"Infosys Springboard."

Submitted to

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

for fulfillment of the requirement for the degree of

BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE & ENGINEERING

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MAHATMA GANDHI MISSION'S COLLEGE OF ENGINEERING
NANDED (M.S.)

Academic Year 2025-26

Certificate

This is to certify that the Internship entitled

“Development of a Dynamic Ride-Sharing and Carpooling Platform”

&

“Infosys Springboard.”

*being submitted by **Mr. Rajsingh Ganeshsingh Thakur** to the Dr. Babasaheb Ambedkar Technological University, Lonere , for the award of the degree of Bachelor of Technology in Computer Science and Engineering, is a record of bonafide work carried out by him under my supervision and guidance. The matter contained in this report has not been submitted to any other university or institute for the award of any degree.*

Ms. Savita S. Wagre

Project Guide

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Computer Science & Engineering

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Infosys Springboard Virtual Internship 6.0 (Batch 13)

“Internship Emails”

Thakur Rajsingh Ganeshsingh <s22_thakur_rajsingh@mgmcen.ac.in>

Springboard Internship 6.0 | Project details!

Infosys Springboard-Support <Springboard-support@infosys.com> Fri, Feb 13, 2026 at 2:32 PM
To: "s22_thakur_rajsingh@mgmcen.ac.in" <s22_thakur_rajsingh@mgmcen.ac.in>

Dear Learner,

With reference to the **Infosys Springboard Internship 6.0**, you should have now received your project title and meeting details. Please attend all scheduled meetings as per the communicated plan. Ensure that you attend every session and complete your assigned tasks on time. **If you have not received any communication, kindly reach out to your mentor**.

Do not miss this valuable opportunity to complete your virtual internship.

If you need to take leave for academic exams or placement activities, please obtain prior written approval via email.

Project title: Development of a Dynamic Ride-Sharing and Carpooling Platform

Mentor email ID: springboardmentor211@gmail.com

Got a query? Write to **springboard-support@infosys.com**

Regards,
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Infosys Springboard Virtual Internship 6.0 (Batch 13)

“Internship Emails”

Thakur Rajsingh Ganeshsingh <s22_thakur_rajsingh@mgmcen.ac.in>

Congratulations on successfully completing the Infosys Springboard Internship 6.0

Infosys Springboard-Support <Springboard-support@infosys.com> Wed, Apr 29, 2026 at 3:29 PM

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Your certificate will be sent to your registered email address shortly. Keep up the great work and continue striving for excellence!

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Note: Your project is your asset. Refrain from sharing your artifacts in online forums/domains to safeguard your work's ownership

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for actively participating and completing the mandatory assignment related to

Internship 6.0 (B 13): Development of a Dynamic Ride Sharing and Carpooling Platform

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With Deep Reverence,

Rajsingh Ganeshsingh Thakur

[B.Tech CSE-A]

ABSTRACT

The project titled Development of a Dynamic Ride-Sharing and Carpooling Platform focuses on creating a web-based transportation system called Smart Ride Sharing. The platform allows drivers and passengers traveling in the same direction to share rides conveniently. Drivers can publish their trip details such as source, destination, travel date, and available seats, while passengers can search for suitable rides and book seats according to their travel requirements. The main objective of the Smart Ride Sharing website is to make travel more efficient, economical, and accessible.

The Smart Ride Sharing platform includes several core modules such as user registration and authentication, ride posting and booking, and route-based ride matching. The system also implements dynamic fare calculation based on the distance traveled by each passenger. To enhance usability, the platform provides booking confirmations, real-time notifications, and a secure online payment system that ensures smooth and reliable transactions between drivers and passengers.

Additionally, the platform contains an administrative dashboard to monitor ride activities, manage users, and track payments within the system. A review and rating feature is also included to maintain transparency and trust between drivers and passengers. By promoting shared transportation through the Smart Ride Sharing website, the system aims to reduce traffic congestion, minimize travel costs, and contribute to a more sustainable and eco-friendly transportation environment.

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INTRODUCTION

Internships play an important role in bridging the gap between academic learning and industry requirements by providing practical exposure to real-world work environments and modern technologies. As part of my B.Tech final year, I participated in the Infosys Springboard Virtual Internship 6.0 (Batch 13) as a Java Full Stack Developer Intern, where I worked on the project "Development of a Dynamic Ride-Sharing and Carpooling Platform." The internship provided hands-on experience in full-stack web application development using technologies such as Java Spring Boot, React.js, MySQL, REST APIs, JWT Authentication, and JPA Hibernate ORM. Through this opportunity, I gained valuable knowledge of software development processes, project implementation methodologies, problem-solving techniques, and industry-oriented development practices, which strengthened my technical skills and enhanced my readiness for professional software engineering roles.

1.1 Internship Overview

The Infosys Springboard Virtual Internship 6.0 was designed to provide practical exposure to students through industry-oriented project development and technical learning activities. As a Java Full Stack Developer intern, I gained hands-on experience in backend development, frontend development, database management, and application integration. The internship focused on improving technical competencies and understanding the workflow involved in real-world software development projects. During the internship, I learned about various stages of software development including requirement analysis, system design, coding, testing, debugging, and project management. The internship also helped me understand professional development practices, teamwork coordination, and the implementation of modern web technologies. Through continuous learning and practical execution, I was able to improve my technical confidence and gain better knowledge of industry standards and development methodologies.

As part of this internship, I successfully completed a project titled "Dynamic Ride Sharing and Carpooling Platform." The primary objective of this project was to develop a web-based system that connects drivers and passengers traveling in the same direction. The platform was designed to reduce traffic congestion, minimize transportation expenses, and support environmentally sustainable travel solutions through shared mobility. The Smart Ride Sharing System allows drivers to publish ride details while passengers can search and book available rides in real time. The application provides features such as ride booking, fare calculation, payment tracking, and user notifications to ensure convenience and transparency for users. The project supports modern smart transportation concepts by encouraging ride sharing and optimizing transportation resources.

The project was developed using modern technologies including Java Spring Boot for backend development, MySQL for database management, and HTML, CSS, JavaScript, and React.js for frontend development. The application follows a modular architecture where separate modules handle specific functionalities such as user management, ride booking, payment processing, fare calculation, and notifications. This modular approach improved the efficiency, scalability, and maintainability of the system. The development process started with requirement analysis and system planning, followed by database design and application architecture preparation. Core functionalities such as user authentication, ride posting, ride searching, and booking management were implemented during the development phase. Dynamic fare calculation was included to automatically determine ride costs based on travel distance and related parameters. Notification features were also integrated to provide updates regarding ride confirmations, booking status, and reminders.

An Admin Dashboard was additionally developed to allow administrators to monitor users, rides, bookings, and overall system activities efficiently. This feature improved administrative management and system monitoring capabilities. The project demonstrates the practical implementation of Java Full Stack Development concepts and highlights the role of technology in solving modern transportation challenges. The successful completion of the Infosys Springboard Virtual Internship 6.0 significantly enhanced my technical knowledge, practical development skills, and understanding of full-stack web application development. This internship experience also improved my confidence in handling real-world projects and prepared me for future professional opportunities in the software development industry.

1.2 Company Profile

Infosys is a globally recognized information technology and consulting company known for delivering innovative digital solutions and advanced software services to organizations across the world. Established in 1981 by N. R. Narayana Murthy and his co-founders, the company has grown into one of the most trusted IT service providers in the technology industry. Infosys operates in multiple sectors including banking, healthcare, retail, education, manufacturing, and telecommunications by offering reliable and modern technology-driven solutions.

Fig. 1.1: Company Logo

Infosys Springboard is the digital learning and skill development initiative launched by Infosys to support students, educators, and professionals with industry-oriented education and training opportunities. The platform provides updated and high-quality learning resources in areas such as Java Full Stack Development, Artificial Intelligence, Machine Learning, Cloud Computing, Data Analytics, Cybersecurity, Web Development, and professional soft skills. Through certification programs, virtual internships, project-based learning, and technical assessments, Infosys Springboard helps learners improve their practical knowledge and stay updated with current industry technologies and trends. The company continuously focuses on innovation, digital transformation, and advanced technology services to meet modern industry requirements. Infosys provides updated solutions in cloud services, automation, enterprise software development, cybersecurity management, AI-driven applications, and consulting services for global businesses. The organization is also dedicated to empowering students and young professionals by offering modern internship programs, technical training, and real-world project experience that help learners become industry-ready and professionally skilled.

1.3 Skill Development

During the internship period, I enhanced my practical knowledge in Java Full Stack Development by working on a real-time Smart Ride Sharing and Carpooling Platform. The internship provided me with hands-on exposure to industry-level software development practices and helped me understand how modern web applications are designed, developed, and managed. It also improved my problem-solving abilities, logical thinking, and technical implementation skills. For backend development, I worked with Java Spring Boot to build scalable and efficient server-side functionalities. Through this framework, I learned how to develop REST APIs, manage application workflows, and handle business logic effectively. I also gained experience with the MVC (Model-View-Controller) architecture, which helped in organizing the project structure in a clean and maintainable manner for better application performance and code management.

The project also helped me improve my database management skills using MySQL. I learned how to create and manage relational databases, store user and ride information, and perform data operations efficiently. In addition, I worked with JPA Hibernate ORM to establish object-relational mapping between Java classes and database tables. This improved my understanding of entity relationships, query execution, data persistence, and transaction handling in enterprise applications. For frontend development, I used HTML, CSS, JavaScript, and React.js to design responsive and interactive user interfaces. React.js helped me understand component-based development, dynamic rendering, and reusable UI structures. I also learned how frontend applications communicate with backend services using REST APIs, enabling smooth data exchange and real-time application functionality.

The internship further strengthened my understanding of application security and project management tools. I implemented JWT (JSON Web Token) authentication with a one-hour session expiration feature, where users are automatically redirected to the login page after token expiry to maintain secure access control. Additionally, I worked with Maven for dependency management and project building, which helped streamline the development process and improve project organization. Overall, the internship significantly improved my technical expertise and practical understanding of full-stack web application development.

1.4 Projects and Work Experience

During my Infosys Springboard Virtual Internship 6.0, which was part of Batch 13, I was involved in a significant project called "Development of a Dynamic Ride-Sharing and Carpooling Platform." This project was a web-based application designed to connect drivers with passengers traveling in the same direction. The main goal of the system was to reduce traffic, lower travel costs, and promote eco-friendly transportation by enabling shared mobility.

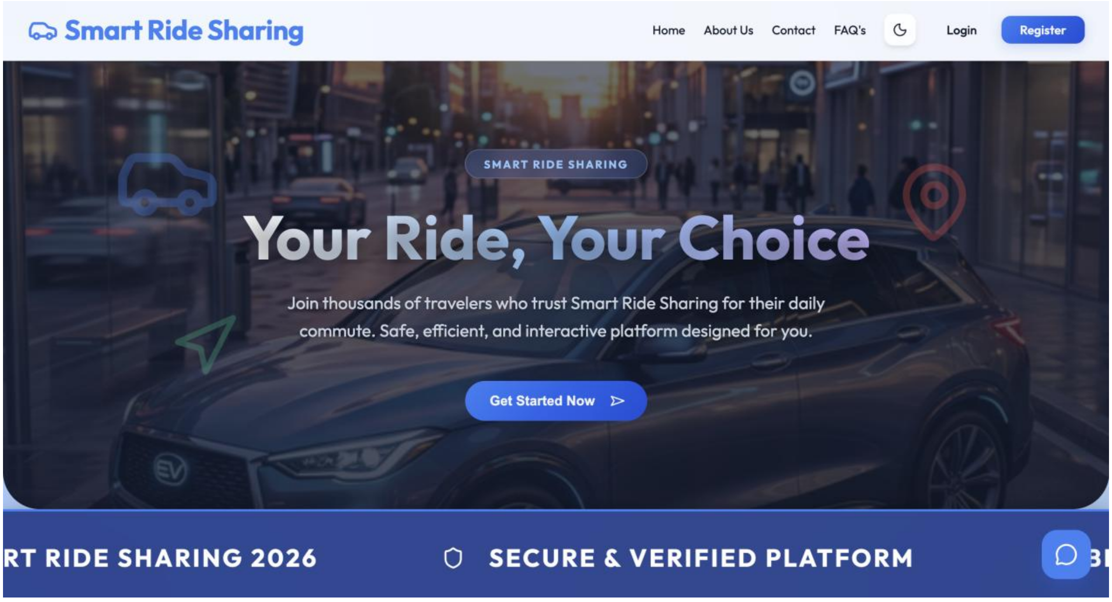


Fig. 1.2: Home Page

As a Java Full Stack Developer intern, I took part in the design, development, and implementation of various parts of the application using current web technologies and development methodologies. The project's homepage was designed with a modern and easy-to-use interface to improve accessibility and user engagement, as shown in Fig. 1.2. Drivers can share their ride details, including origin, destination, travel date, and number of available seats, while passengers can search for and book rides according to their needs. The system includes features such as ride booking, route-based matching, fare calculation, payment tracking, notifications, and user review management. These features were created to ensure ease of use, clarity, and effective user communication.

For the project development, I used Java Spring Boot for the backend, MySQL for database management, and HTML, CSS, JavaScript, and React.js for the frontend.

The application followed the MVC architecture and utilized REST APIs to ensure smooth communication between the frontend and backend systems. I also implemented JWT authentication for secure login with token expiration. Users are redirected to the login page after one hour of inactivity to maintain application security. Additionally, I used Maven for dependency management and JPA Hibernate ORM for efficient database operations and project structure. During the internship, I gained hands-on experience in requirement analysis, database design, backend and frontend development, debugging, testing, and documentation. I contributed to features like dynamic fare calculation, real-time notifications, ride confirmation updates, and an Admin Dashboard to monitor users and rides. This project helped me strengthen my technical skills, improve my teamwork abilities, and deepen my understanding of full-stack web development. Overall, the internship provided valuable industry experience and boosted my confidence in dealing with real-world software development challenges.

1.5 Learning Outcomes

During the Infosys Springboard Virtual Internship 6.0, I gained valuable practical experience in Java Full Stack Development and improved my understanding of real-world software development processes. The internship helped me strengthen my technical knowledge in backend development using Java Spring Boot, frontend development using React.js, and database management using MySQL. I also learned how to develop REST APIs, implement MVC architecture, manage dependencies using Maven, and perform database operations using JPA Hibernate ORM. Working on a real-time project improved my coding skills, debugging abilities, logical thinking, and understanding of application development workflows. The internship also enhanced my knowledge of professional development practices, teamwork, project planning, and system implementation. Through the Smart Ride Sharing and Carpooling Platform project, I learned how to design secure and user-friendly applications with features such as JWT authentication, ride booking, dynamic fare calculation, notifications, and admin management systems. Additionally, I gained experience in testing, problem-solving, and integrating frontend and backend technologies efficiently. Overall, this internship improved my confidence, technical expertise, and readiness to work on industry-level software projects in the future.

1.6 Organization of the Report

This report is structured into five chapters:

Chapter 1: Introduction - Presents the internship overview, company profile, skill development activities, project experience, and key learning outcomes achieved during the internship period.

Chapter 2: Company Background - Describes Infosys Springboard, its learning initiatives, certification programs, virtual internship opportunities, technical training resources, and skill development activities offered to students and learners. Describes the company's services, vision, mission, certifications, and community initiatives like Skill-Building Meetups.

Chapter 3: Technology Stack - Explains the technologies, frameworks, development tools, authentication mechanisms, database technologies, and build management tools used during the development of the Smart Ride Sharing System.

Chapter 4: System Architecture and Task Assignment - Discusses the proposed system architecture, workflow design, application layers, project modules, milestone-wise task allocation, and development planning followed throughout the internship project.

Chapter 5: Internship Project Development and Implementation - Provides a detailed description of the Smart Ride Sharing System, including requirement analysis, system design, frontend and backend development, database implementation, ride posting and booking functionalities, payment integration, notification services, testing activities, challenges encountered, and project outcomes achieved during the internship.

Chapter 2

COMPANY BACKGROUND

Infosys Springboard is an online learning and skill development platform that provides students, educators, and professionals with access to quality educational resources, certification programs, virtual internships, and industry-focused training opportunities. The platform is designed to enhance practical knowledge and bridge the gap between academic learning and real-world industry requirements through project-based learning, assessments, coding exercises, and hands-on experiences. It offers training in various domains such as programming, web development, artificial intelligence, cloud computing, cybersecurity, data science, digital technologies, and professional skills. By promoting self-paced learning, continuous upskilling, and practical implementation of concepts, Infosys Springboard helps learners build technical expertise, improve problem-solving abilities, and develop industry-ready competencies that support their academic, professional, and career growth.

2.1 Services Provided

Infosys Springboard provides a variety of educational and technical learning services for students, fresh graduates, teachers, and working professionals. The platform offers online certification courses, virtual internship programs, project-based learning activities, coding practice sessions, and technical assessments in multiple technology domains. These services help learners improve their practical understanding and gain knowledge aligned with current industry standards. As shown in Fig. 2.1. The platform delivers updated learning programs in fields such as Java Full Stack Development, Artificial Intelligence, Machine Learning, Cloud Computing, Cybersecurity, Data Analytics, Web Development, and Digital Technologies. In addition to technical training, Infosys Springboard also provides soft skill development programs that focus on communication, leadership, teamwork, problem-solving, and professional ethics. These services support the overall development of learners and prepare them for real-world professional environments. Infosys Springboard additionally provides practical exposure through hands-on projects and virtual internship opportunities where learners can work on real-time applications and industry-oriented problem statements.

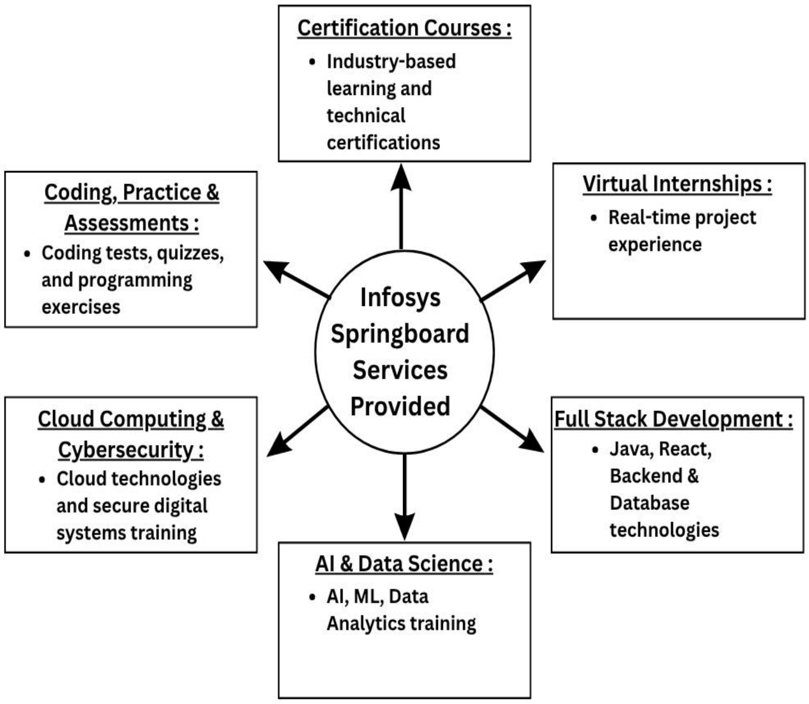


Fig. 2.1: Services Provided

The platform also includes learning resources such as recorded lectures, quizzes, coding challenges, and interactive study materials that help users strengthen their technical and analytical skills. By continuously updating its courses and learning content, the platform ensures that learners remain connected with modern technologies and industry requirements.

2.1.1 Certification Courses

Infosys Springboard offers a broad selection of certification courses aimed at helping students and professionals enhance their technical and professional knowledge. These programs are created based on current industry standards and cover areas like programming, web development, artificial intelligence, cloud computing, cybersecurity, data analytics, and digital technologies. The platform provides organized learning materials, recorded lessons, assessments, and hands-on activities that support both understanding of theory and development of technical skills. These certification courses assist learners in strengthening their academic background and increasing their career prospects by gaining recognized technical knowledge and practical experience. These programs also promote self-paced learning, allowing students to access educational resources at their own convenience and according to their learning pace. Through ongoing learning and completing certifications, learners can boost their confidence, technical abilities, and readiness for future opportunities in the technology field.

2.1.2 Virtual Internship Programs

Infosys Springboard provides virtual internship programs that offer students real-world experience and exposure to industry projects. These internships enable learners to understand software development processes, project implementation strategies, teamwork, and professional practices. Participants work on industry-focused problems and develop hands-on experience in designing, building, testing, and managing technical applications. The internship programs also help students improve their communication skills, problem-solving abilities, understanding of project management, and technical implementation knowledge. Learners are encouraged to apply their theoretical knowledge in practical settings while working with modern technologies and development practices. These internships create opportunities for students to gain industry experience and prepare for professional roles in software development and information technology.

2.1.3 Full Stack Development Training

Infosys Springboard delivers Full Stack Development training programs that focus on both frontend and backend technologies necessary for developing modern web applications. The platform offers learning content and practical guidance on technologies such as Java, Spring Boot, React.js, HTML, CSS, JavaScript, MySQL, REST APIs, and database management systems. These training programs help learners understand the entire application development process, from designing user interfaces to implementing backend server solutions. In addition, the Full Stack Development training helps learners improve coding practices, debugging skills, knowledge of API integration, and techniques for handling databases. Through project-based learning and practical implementation activities, students gain experience in building secure, responsive, and scalable web applications. The programs also enhance learners' understanding of software architecture, system design, and real-world application development practices.

2.1.4 Cloud Computing & Cybersecurity

Infosys Springboard provides learning opportunities in Cloud Computing and Cybersecurity to help learners understand modern digital infrastructure and secure computing environments. The platform offers educational resources covering topics like cloud services, data security, network protection, ethical security practices, and secure application management. These programs help students gain awareness of current technological advancements and security challenges in the digital world. The Cloud Computing and Cybersecurity programs also support learners in understanding safe data handling, digital privacy, secure communication systems, and cloud-based technology solutions. Practical learning activities and technical content help students develop analytical thinking and problem-solving skills related to system security and cloud technologies. These programs prepare learners for emerging career opportunities in cybersecurity and cloud technology fields.

2.1.5 Artificial Intelligence & Data Science

Infosys Springboard offers specialized learning programs in Artificial Intelligence, Machine Learning, and Data Science to help learners understand advanced technologies and intelligent digital systems. These programs include training on data analysis, predictive modeling, machine learning algorithms, AI-based applications, and modern computational techniques. Learners gain insight into how intelligent systems are developed and applied in various industries for automation and decision-making. The Artificial Intelligence and Data Science learning programs also help students improve logical reasoning, analytical thinking, and data interpretation skills. Through practical exercises, technical assignments, and project-based learning activities, learners understand real-world applications of AI and data-driven technologies. These programs support continuous learning and help students stay informed about emerging technological trends and innovations.

2.1.6 Coding Practice & Assessments

Infosys Springboard includes coding practice sessions, quizzes, technical assessments, and project-based activities that help learners build and enhance their programming and logical reasoning skills. The coding exercises cover various programming languages and technical concepts, enabling students to improve their problem-solving abilities and practical coding knowledge. These learning activities support ongoing practice and help learners gain confidence in software development and technical implementation. The assessment programs also help learners evaluate their technical understanding, analytical thinking, and coding efficiency through consistent practice and performance tracking. Coding challenges and technical tasks encourage learners to apply theoretical concepts in real situations and progressively improve their development skills. Through these interactive educational activities, learners gain better preparation for technical interviews, project development, and professional software engineering roles.

2.2 Vision, Mission & Certifications

The vision of Infosys Springboard is to create equal learning opportunities for students and learners by providing accessible, high-quality, and industry-relevant education through digital platforms. The initiative focuses on empowering individuals with technical knowledge, professional skills, and practical experience that can help them build successful careers in the technology and digital sectors. The mission of Infosys Springboard is to support learners in developing future-ready skills through innovative learning methods, certification programs, virtual internships, and practical project experiences. The platform is committed to promoting continuous learning, technical excellence, creativity, and professional growth by making modern educational resources available to a wide range of learners. It also aims to encourage skill development that aligns with evolving industry demands and technological advancements.

Infosys Springboard follows quality-driven educational practices and is supported by globally recognized organizational standards associated with professional learning, information security, and quality management systems. The platform operates within an environment that emphasizes secure digital learning, updated educational practices, and professional training standards. It is associated with internationally recognized certifications and standards such as ISO 9001 for Quality Management Systems and ISO 27001 for Information Security Management Systems, which reflect a commitment toward maintaining quality learning services, secure digital operations, and reliable educational support for learners worldwide.

2.3 Community Engagement & Skill-Building Meetups

Infosys Springboard promotes community engagement by creating a collaborative learning environment where students, educators, and professionals can interact and share knowledge. The platform encourages learners to participate in various technical discussions, group learning activities, coding practices, and knowledge-sharing sessions that help improve both technical and communication skills. These community-driven initiatives help learners build confidence, teamwork abilities, and professional connections while learning modern technologies. Skill-Building Meetups organized through Infosys Springboard provide opportunities for learners to interact with mentors, industry professionals, and fellow participants through virtual events and technical sessions. These meetups include webinars, workshops, expert talks, project presentations, and career guidance sessions that help learners gain awareness about current industry trends and technological advancements. Such interactive programs support continuous learning and motivate students to explore innovative ideas and practical solutions.

Through these engagement activities, participants are able to enhance their problem-solving skills, leadership qualities, and collaborative working abilities. Learners also get opportunities to discuss technical challenges, share project experiences, and receive guidance from experienced professionals. The platform creates a supportive learning community where students can improve their practical understanding and gain exposure to real-world industry practices. Community engagement programs and Skill-Building Meetups additionally contribute to the overall professional development of learners by encouraging active participation, networking, and continuous skill improvement. These activities help students remain updated with emerging technologies and industry requirements while developing confidence in presenting ideas and working in collaborative environments. Overall, Infosys Springboard’s community initiatives play an important role in strengthening technical learning, professional growth, and career readiness among participants.

Chapter 3

TECHNOLOGY STACK

The Smart Ride Sharing System was developed as a full-stack web application to provide a secure and efficient platform for ride-sharing between drivers and passengers. The frontend was built using HTML, CSS, JavaScript, and React.js, while the backend was implemented using Java Spring Boot following the MVC architecture and REST API-based communication. MySQL and JPA Hibernate ORM were used for data management, and technologies such as JWT Authentication, Maven, and Stripe Payment Integration enhanced application security, dependency management, and online payment processing. The system successfully integrated ride booking, fare calculation, notifications, and user management functionalities to deliver a reliable and user-friendly transportation solution.

3.1 Frontend Technologies

The frontend section of the Smart Ride Sharing System was developed using HTML, CSS, JavaScript, and React.js. HTML was used for structuring web pages, while CSS was used for styling and improving the visual appearance of the application. JavaScript was implemented to add dynamic functionalities and interactive behavior to the user interface. React.js was used as the primary frontend framework because of its component-based architecture and efficient rendering capabilities. It helped in creating reusable UI components, improving application responsiveness, and enabling smooth interaction between frontend pages and backend services. The frontend interface was designed to provide better accessibility, responsiveness, and user experience for both drivers and passengers.

The frontend technologies played an important role in creating a modern and visually attractive interface for the application. Different UI components such as navigation bars, booking forms, ride search panels, and dashboards were designed to provide users with easy navigation and improved usability. Responsive web design techniques were also applied to ensure compatibility across different devices and screen sizes. React.js additionally improved the overall efficiency of the application by enabling faster rendering and simplified state management. The use of reusable components reduced code duplication and improved maintainability of the frontend system. Through frontend development, the project achieved a user-friendly environment that supports smooth interaction between passengers, drivers, and administrators.

3.1.1 Key Features of Frontend Technologies

The frontend technologies used in the Smart Ride Sharing System mainly included HTML, CSS, JavaScript, and React.js. These technologies helped in creating an interactive, responsive, and visually appealing user interface for the application. HTML was used to structure the web pages, while CSS improved the design, layout, and presentation of the platform. JavaScript provided dynamic functionalities that enhanced user interaction and application responsiveness. React.js was used as the main frontend framework because of its component-based architecture and efficient rendering capabilities. It allowed the development of reusable components, which improved code organization and reduced development complexity. React.js also helped in managing application states and enabled faster updates within the user interface without reloading the complete web page. The frontend technologies supported responsive web design and smooth navigation for drivers, passengers, and administrators. Different UI sections such as ride search forms, booking panels, user dashboards, and notification interfaces were designed for better accessibility and user convenience. These frontend features improved the overall usability and user experience of the ride-sharing platform.

3.1.2 Use Case in My Internship

During my internship, frontend technologies were used to design and implement the user interface of the Smart Ride Sharing System. I worked on creating responsive web pages that allowed users to register, log in, search for rides, post ride details, and book available seats. The interface was designed to ensure smooth interaction between users and the application. React.js was used to develop reusable components for different modules such as ride booking, user management, and notifications. JavaScript functionalities were implemented to provide dynamic behavior and real-time updates within the application. CSS styling techniques were also applied to improve the appearance and responsiveness of the platform across multiple devices.

Frontend technologies played an important role in improving accessibility and user engagement during the project development phase. The homepage, booking sections, and user dashboard interfaces were designed to provide clear navigation and efficient user interaction. These implementations helped create a modern and user-friendly ride-sharing application during the internship project.

3.2 Backend Technologies

The backend development of the project was implemented using Java Spring Boot. This framework was used to develop server-side functionalities, handle application logic, and manage API communication. Spring Boot simplified the development process by providing built-in configurations and tools for creating scalable enterprise-level applications. The project also followed the MVC (Model-View-Controller) architecture, which separated application logic into different layers for better organization and maintainability. REST APIs were developed to enable smooth communication between frontend and backend modules for functionalities such as ride booking, user authentication, notifications, and fare calculation.

Fig. 3.1: Java Spring Boot Logo

As shown in Fig. 3.1. Java Spring Boot provided a reliable environment for implementing business logic and handling user requests efficiently. It supported secure data processing, API integration, and smooth management of application workflows. The backend system handled important functionalities such as ride posting, booking confirmation, fare calculation, user management, and notification processing. The MVC architecture improved the modular structure of the project by separating the application into models, views, and controllers. This separation enhanced code readability, maintainability, and scalability of the application. REST API implementation also helped establish secure and efficient communication between frontend and backend systems for real-time operations.

3.2.1 Key Features of Backend Technologies

Java Spring Boot was used as the primary backend technology for developing the Smart Ride Sharing System. It provided a reliable framework for handling server-side functionalities, business logic implementation, and API communication. Spring Boot simplified application development by offering built-in configurations, dependency management, and enterprise-level support. The backend system followed the MVC (Model-View-Controller) architecture, which separated application functionalities into models, views, and controllers. This structure improved project organization, maintainability, scalability, and readability of the application. REST APIs were also implemented for secure communication between frontend and backend modules. Backend technologies additionally supported functionalities such as ride posting, booking confirmation, notification management, user authentication, fare calculation, and payment tracking. The backend system ensured smooth processing of user requests and maintained proper workflow management throughout the ride-sharing application.

3.2.2 Use Case in My Internship

During the internship project, Java Spring Boot was used to develop the core backend functionalities of the Smart Ride Sharing System. I worked on implementing APIs for ride management, booking operations, user authentication, and notification handling. The backend logic helped process user requests efficiently and ensured proper communication between the frontend and database systems. The MVC architecture was applied to organize the project into separate layers for better maintainability and structured development. Controllers managed incoming requests, service layers handled business logic, and repository layers interacted with the database system. This architecture improved code organization and simplified debugging during development. REST APIs were implemented for functionalities such as ride booking, dynamic fare calculation, payment processing, and route-based ride management. These APIs enabled secure data exchange between the frontend interface and backend server, ensuring real-time updates and smooth application performance during project execution.

3.3 Database Technologies

MySQL was used as the relational database management system for storing and organizing application data securely and efficiently. The database stored information related to users, rides, bookings, payments, reviews, and notifications. MySQL helped in managing structured data and ensured reliable data handling throughout the application as shown in Fig. 3.2. JPA Hibernate ORM was implemented to establish object-relational mapping between Java classes and database tables. This technology simplified database interaction by reducing manual SQL query handling and improving data persistence operations. Hibernate also improved development efficiency and maintainability of the application.

Fig. 3.2: MySQL Workbench Logo

The database system was designed to manage large amounts of application data in an organized and secure manner. Proper relationships were created between database tables to maintain data consistency and improve retrieval performance. MySQL also supported efficient query execution and transaction handling for ride booking and payment management functionalities. JPA Hibernate ORM reduced the complexity of database connectivity by automating CRUD operations and object mapping processes. It enabled smoother interaction between backend services and the database layer while improving development speed. This technology also helped in maintaining clean and structured database operations throughout the project lifecycle.

3.3.1 Key Features of Database Technologies

MySQL was used as the relational database management system for storing and managing application data securely and efficiently. The database handled important information related to users, rides, bookings, payments, notifications, and reviews. MySQL supported structured data storage and reliable transaction management within the application. JPA Hibernate ORM was implemented to establish object-relational mapping between Java classes and database tables. This technology reduced the need for manual SQL query handling and simplified database operations through automated persistence management. Hibernate also improved data consistency and reduced development complexity. Database technologies provided secure data management, faster query execution, and efficient handling of application records. Proper database relationships and constraints were implemented to maintain data integrity and support smooth ride booking and transaction management functionalities within the system.

3.3.2 Use Case in My Internship

During the internship project, MySQL was used to manage all application-related data such as user profiles, ride details, booking information, payment records, and notifications. The database structure was designed to support efficient data retrieval and secure information storage for different modules of the application. JPA Hibernate ORM was used to connect backend services with the database layer and automate CRUD operations. It simplified interaction between Java entities and database tables, reducing manual database coding efforts during project development. Hibernate also improved maintainability and performance of database operations. Database technologies were mainly applied in modules such as user management, ride booking, transaction history, and notification storage. These implementations ensured reliable data processing and efficient management of real-time information within the Smart Ride Sharing System during the internship phase.

3.4 Authentication & Security Technologies

JWT (JSON Web Token) Authentication was implemented in the Smart Ride Sharing System to provide secure login access and session management for drivers, passengers, and administrators. The token-based authentication mechanism ensured that only authorized users could access application functionalities while protecting sensitive user information from unauthorized access. Secure token generation and verification additionally improved communication between frontend and backend services throughout the application workflow. The system also included a one-hour token expiration feature where users are automatically redirected to the login page after session expiry to improve application security. Additional security measures were implemented to protect booking records, payment details, and user profile information through secure authentication and session handling practices. These security technologies improved reliability, confidentiality, and overall trust within the Smart Ride Sharing System.

3.4.1 Key Features of Authentication & Security Technologies

JWT (JSON Web Token) Authentication was used in the project to provide secure user login and session management functionalities. Token-based authentication ensured that only authorized users could access protected modules and application resources. This technology improved security and prevented unauthorized access within the platform. JWT authentication provided secure token generation and verification mechanisms for validating user sessions. The authentication process helped maintain secure communication between frontend and backend systems by transmitting encrypted tokens instead of exposing sensitive user credentials repeatedly. The project also included session expiration functionality for better application security. Tokens were configured with a one-hour expiration period, after which users were automatically redirected to the login page. This feature helped maintain secure session control and protected user data from misuse or unauthorized access. Additionally, role-based access control was implemented to ensure that users could only access features and functionalities permitted according to their assigned roles within the system.

3.4.2 Use Case in My Internship

During my internship, JWT authentication was implemented to secure user login and access control within the Smart Ride Sharing System. The authentication mechanism validated user identities during login and generated secure tokens for maintaining authenticated sessions during application usage. The security implementation was mainly applied to protect modules such as ride posting, booking management, payment tracking, and user profile management. Only authenticated users were allowed to access authorized functionalities, which improved the reliability and security of the platform. The one-hour token expiration functionality was also implemented during project development to improve session management and application security. After token expiry, users were redirected to the login page to re-authenticate themselves. This security implementation enhanced safe access control and secure application usage during the internship project.

3.5 Development Tools & Build Management

Maven was used as the project build and dependency management tool during the development process. It helped manage required libraries, dependencies, plugins, and project configurations efficiently. Maven simplified the build process and improved project organization for smoother development and deployment activities. Additional development tools and debugging practices were also used for testing, application monitoring, and code management throughout the internship project. These tools helped improve code quality, maintainability, and overall project performance while supporting efficient full-stack web application development.

Maven provided a centralized approach for handling project dependencies and ensured compatibility between different libraries used in the application. It also automated tasks such as project compilation, packaging, and dependency updates, which reduced manual effort and improved development productivity. The use of Maven helped maintain a well-structured and manageable project environment. Different debugging and testing methods were applied throughout the project development phase to identify and resolve errors effectively. Code optimization, testing procedures, and monitoring techniques helped improve application stability and overall system performance. These development practices contributed to creating a reliable and efficient web-based ride-sharing platform.

Development Tools and Build Management played an important role in the successful implementation of the Smart Ride Sharing System by supporting efficient coding, dependency management, project configuration, and application deployment. Maven was used as the build automation and dependency management tool to handle external libraries, simplify project setup, and maintain consistent development environments. Development activities were carried out using modern programming tools and frameworks that improved code organization, version control, debugging, testing, and overall productivity. These tools helped streamline the development process, reduce manual configuration efforts, and ensure reliable application performance throughout the project lifecycle.

3.5.1 Key Features of Development Tools & Build Management

Maven was used as the primary build and dependency management tool for the Smart Ride Sharing System project. It helped manage project dependencies, plugins, and required libraries efficiently through a centralized configuration structure. Maven simplified the build process and improved overall project organization. The tool automated various development tasks such as compilation, packaging, dependency installation, and project configuration management. Maven also reduced manual dependency handling and ensured compatibility between different libraries and frameworks used in the application. Development tools and debugging practices further supported testing, code management, and project monitoring throughout the development process. These tools improved application stability, maintainability, and development productivity while supporting efficient software engineering practices.

3.5.2 Use Case in My Internship

During the internship project, Maven was used to manage all required project dependencies and framework libraries for frontend and backend development. It simplified dependency installation and helped maintain a properly structured development environment throughout the project lifecycle. Maven was also used for project building, compilation, and packaging activities during application development and testing phases. The automated build management process reduced development complexity and improved workflow efficiency while working on multiple project modules. Additional debugging and testing practices were applied to identify errors, improve code quality, and maintain application performance. These tools and development methodologies helped ensure smooth execution of the Smart Ride Sharing System and supported successful completion of the internship project.

Various development and support tools were also utilized to improve productivity, maintain code quality, and streamline the software development process. Version control practices helped manage project updates efficiently, while integrated development environments (IDEs) provided features such as code suggestions, debugging support, and project navigation. Regular testing, error tracking, and performance optimization activities were carried out throughout development to ensure application stability, reliability, and maintainability. The effective use of these tools contributed to a more organized development workflow and enhanced the overall quality of the Smart Ride Sharing System.

Chapter 4

SYSTEM ARCHITECTURE AND TASK ASSIGNMENT

The System Architecture and Task Assignment of the Smart Ride Sharing System were designed to ensure organized development and efficient implementation of the project. The architecture followed a layered approach consisting of Presentation, Authentication, Application, Admin, and Database layers, which enabled secure communication, smooth data flow, and effective management of ride-sharing operations. The project tasks were systematically divided into four milestones covering user management, ride posting and booking, payment integration, notification and review management, testing, and documentation. This structured approach supported successful project execution, improved development efficiency, and provided valuable practical experience in full-stack web application development and real-world software engineering practices.

4.1 Overview of Proposed System and Workflow

The Smart Ride Sharing System was designed as a web-based transportation platform that enables drivers and passengers traveling in the same direction to connect through a secure and efficient ride-sharing environment. The system architecture follows a layered approach where each layer performs specific responsibilities such as user interaction, authentication, business logic processing, administration, and database management. This architecture improves system organization, simplifies module integration, and ensures reliable communication between frontend, backend, payment systems, and database services. The workflow of the proposed system begins with driver and passenger registration through the web interface. After successful login and authentication, drivers can post ride details such as source location, destination, travel date, seat availability, and travel preferences. Passengers can search available rides according to their destination and reserve seats based on ride availability and route matching. Once the booking process is completed, the system performs dynamic fare calculation and secure Stripe payment processing for online transactions. The application additionally provides notification functionalities for booking confirmations, ride reminders, payment updates, and ride status information. After successful ride completion, passengers and drivers can submit reviews and ratings through the platform. The entire system is monitored through an admin dashboard, while all user records, ride information, booking details, payment transactions, and reviews are securely stored within the database layer. This architecture supports smooth ride-sharing operations, secure communication, and better transportation management.

4.1.1 Presentation Layer

The Presentation Layer acts as the user interaction layer of the Smart Ride Sharing System and provides the graphical interface through which drivers, passengers, and administrators access application functionalities. This layer was developed using React.js along with HTML, CSS, and JavaScript to create a responsive, interactive, and user-friendly web interface. The frontend design was created to improve accessibility, navigation, and overall user experience across different devices and screen sizes.

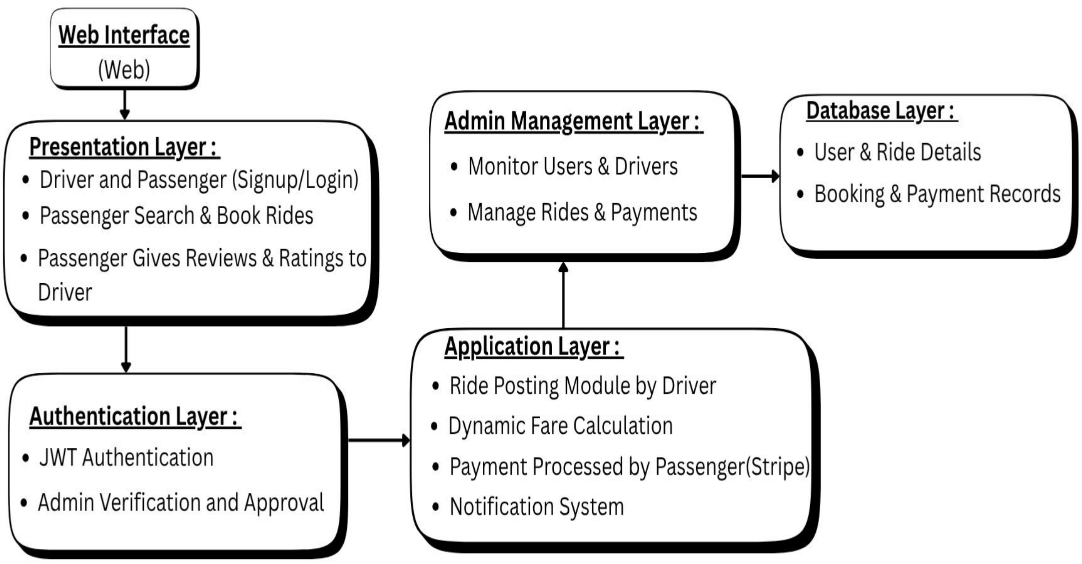


Fig. 4.1: System Architecture

The Presentation Layer allows drivers and passengers to perform important operations such as user registration, secure login, profile management, ride posting, ride searching, ride booking, payment access, and review submission. Drivers can enter ride-related details such as source location, destination, travel date, available seats, and travel preferences, while passengers can search rides according to their travel requirements and reserve available seats through the interface. The layer additionally provides real-time ride information, booking status updates, notifications, and payment-related communication for users. User interface components were designed using reusable frontend structures to improve application maintainability and frontend performance. The Presentation Layer plays an important role in ensuring smooth interaction between users and backend services throughout the ride-sharing workflow.

4.1.2 Authentication Layer

The Authentication Layer was developed to ensure secure login verification, user authentication, and session management in the Smart Ride Sharing System. JWT (JSON Web Token) authentication was incorporated to allow only authorized users to access restricted features of the application. During the login process, the system checks user credentials and issues secure access tokens to authenticated users, thereby enhancing the application's security and preventing unauthorized access to sensitive user data. Additional security features such as password encryption, token validation, and the management of one-hour token expiration were also implemented in this layer. Drivers, passengers, and administrators were granted role-based access permissions according to their designated system privileges. This layer ensured secure interactions between the frontend and backend components while maintaining consistent session control throughout the application process.

4.1.3 Application Layer

The Application Layer was created as the central processing unit of the Smart Ride Sharing System, handling all important business logic and backend tasks. Developed using Java Spring Boot and MVC Architecture, this layer manages features like ride posting, ride booking, route matching, fare calculation, notifications, review management, and secure payment processing. REST APIs were also introduced to support seamless communication between frontend interfaces and backend services. This layer also processes user requests, validates ride and booking details, and interacts with the database layer for secure data storage and retrieval. Integration with Stripe for card payments, transaction tracking, notification services, and dynamic fare calculation were also handled within this layer. This implementation enhanced the application's scalability, maintainability, and the smooth execution of ride-sharing operations across the platform.

4.1.4 Admin Layer

The Admin Layer was implemented to offer centralized monitoring and management tools for the Smart Ride Sharing System. This layer enables administrators to oversee rides, users, bookings, payments, notifications, and reviews through an admin dashboard. The centralized management system improved operational transparency and supported effective administrative control over the platform. The admin dashboard also facilitates the monitoring of payment transactions, ride history, user activities, and system performance. Administrators can identify and resolve issues, manage user accounts, and track application activities throughout the system's lifecycle. This layer contributed to the application's reliability, secure management, and stable operation of the ride-sharing platform.

4.1.5 Database Layer

The Database Layer was designed to securely store and manage all information related to the application within the Smart Ride Sharing System. MySQL was selected as the relational database management system to handle structured data such as user profiles, rides, bookings, transactions, notifications, reviews, and route information. This layer ensured reliable data storage and supported efficient management of all ride-sharing operations within the application. JPA Hibernate ORM was integrated to facilitate communication between Java backend entities and MySQL database tables through object-relational mapping. Backend services interacted with the database layer to perform secure data insertion, retrieval, updating, and deletion. Proper database management improved data consistency, faster query performance, secure storage, and overall application performance within the Smart Ride Sharing System. Database relationships and constraints were carefully defined to maintain data integrity and prevent inconsistencies across interconnected tables. The database layer also supported optimized query execution and indexing techniques, enabling faster data access and improved system responsiveness during high user activity.

Appropriate indexing mechanisms were implemented to enhance query execution speed and optimize system performance.

4.2 Modules to be Implemented

The Smart Ride Sharing System consists of multiple modules designed to handle different functionalities required for secure and efficient ride-sharing operations. Each module performs specific tasks and works together with other modules to maintain smooth communication, reliable data management, and better user experience throughout the platform. The modular architecture improves maintainability, scalability, and system performance within the application.

4.2.1 User Management Module

The User Management Module was developed to handle user registration, login authentication, profile management, and account-related operations for drivers and passengers. This module provides secure access to the platform and manages user information efficiently through authentication and session management functionalities. Both drivers and passengers can create accounts and access application services according to their authorized roles. JWT authentication and secure password handling mechanisms were implemented within this module to improve user security and maintain protected access to application resources. User profiles store important information such as personal details, vehicle information, contact details, travel preferences, and ride history for better ride management and communication. The module additionally supports secure session handling, user verification, and role-based access control for different types of users within the system. Proper account management functionalities improve user accessibility, application security, and efficient management of ride-sharing operations throughout the platform.

4.2.2 Ride Posting and Booking Module

The Ride Posting and Booking Module enables drivers to publish ride details and passengers to search and reserve available rides according to their travel requirements. Drivers can enter ride information such as source location, destination, travel date, departure timing, seat availability, and ride preferences through the application interface. Passengers can search available rides using route-based filtering and booking functionalities. The system displays ride availability and allows passengers to reserve seats based on their preferred travel destination and timing. Booking confirmation functionalities additionally provide ride details, booking updates, and reservation information after successful booking operations. This module improves ride-sharing efficiency and simplifies communication between drivers and passengers throughout the booking process. Real-time ride availability updates and booking management functionalities help maintain smooth transportation coordination and better ride-sharing operations within the application.

4.2.3 Fare Calculation & Payment Module

The Fare Calculation and Payment Module was implemented to manage automatic fare estimation and secure online payment processing within the Smart Ride Sharing System. Dynamic fare calculation was developed to determine ride pricing based on travel distance, ride route, and passenger travel requirements. This functionality improves transparency and ensures fair pricing for both drivers and passengers. The module also provided real-time fare updates, allowing passengers to view estimated ride costs before confirming their bookings.

The fare calculation per seat was performed using the following formula :

Fare Per Seat = Base Fare (in rupees) + (Distance in KM × Rate Per KM (in rupees)) **(1)**

This formula helped calculate ride costs dynamically according to the distance traveled by passengers and ensured accurate fare estimation during ride booking operations. Stripe card payment integration was implemented to support secure online transaction processing through card-based payment methods. Passengers can complete ride payments securely while the system maintains payment history, transaction records, and financial details for future reference and monitoring purposes. The module additionally supports payment tracking, secure transaction handling, and booking confirmation after successful payment processing. Real-time payment updates improve user trust and application reliability while ensuring smooth financial management within the ride-sharing platform.

4.2.4 Route Matching & Tracking Module

The Route Matching and Tracking Module was developed to connect drivers and passengers traveling in the same direction through route-based ride allocation functionalities. The system analyzes travel routes and identifies suitable rides according to passenger destination preferences and driver travel paths. This module improves transportation optimization by reducing unnecessary vehicle usage and promoting efficient shared mobility solutions. Passengers receive real-time ride availability and route-related information while drivers can manage ride coordination efficiently through the application. Ride tracking and route monitoring functionalities additionally improve communication between passengers and drivers throughout the travel process. The module supports better ride coordination, transportation efficiency, and cost-effective travel management within the Smart Ride Sharing System.

4.2.5 Notification and Review System

The Notification and Review System was implemented to provide communication management and feedback functionalities within the application. Users receive notifications related to ride confirmations, booking updates, ride reminders, cancellations, payment alerts, and ride status changes throughout the booking process. The notification module improves coordination between passengers and drivers by providing real-time communication and system-generated alerts. Backend services dynamically generate notifications according to ride activities, payment updates, and booking operations performed within the application. The review management functionality additionally allows passengers and drivers to provide ratings and feedback after successful ride completion. This system improves trust, transparency, accountability, and service quality within the ride-sharing platform by encouraging better user interaction and ride experience evaluation.

4.2.6 Admin Dashboard for Monitoring

The Admin Dashboard Module was implemented to provide centralized supervision and management of all activities within the Smart Ride Sharing System. Through a dedicated dashboard interface, administrators can monitor user accounts, ride details, booking records, payment transactions, reviews, and overall platform performance.

This module helps maintain operational transparency and enables efficient control over different ride-sharing processes. By providing access to important system information in one place, the dashboard simplifies administrative tasks and improves the effectiveness of platform management. The dashboard also supports user verification, passenger approval, ride monitoring, payment tracking, and issue resolution activities. Administrators can review system-generated reports, observe user behavior, identify operational problems, and take necessary actions to ensure smooth application performance. These monitoring capabilities contribute to improved security, better resource management, and reliable service delivery. Overall, the Admin Dashboard plays a significant role in maintaining system stability, enhancing administrative efficiency, and supporting the long-term success of the Smart Ride Sharing System.

4.3 Task Assignment

The internship project titled "Development of a Dynamic Ride-Sharing and Carpooling Platform" was divided into four major milestones to ensure proper planning, systematic implementation, and successful completion of the Smart Ride Sharing System. Each milestone focused on developing important modules and functionalities required for building a secure, efficient, and user-friendly web application. The milestone-based workflow helped in understanding real-world software development processes such as requirement analysis, module implementation, testing, debugging, deployment, and documentation. During the internship, I actively participated in frontend development, backend implementation, database management, API integration, authentication handling, testing, and project documentation activities.

The project development process followed a structured approach where each milestone contributed toward improving the functionality and performance of the Smart Ride Sharing System. Different modules were integrated gradually to ensure smooth communication between frontend, backend, database, payment systems, and notification services. This organized development methodology improved project maintainability, reduced implementation complexity, and supported successful completion of all required project objectives during the internship phase.

4.3.1 Milestone–1 : Management & Ride Posting Module

The first milestone mainly focused on implementing the foundational functionalities of the Smart Ride Sharing System, including user management, authentication, and ride posting operations. During this phase, user registration and login modules were developed for both drivers and passengers to allow secure access to the platform. Secure password management and authentication mechanisms were implemented to maintain user privacy and protect account information from unauthorized access. JWT authentication and secure session handling techniques were also integrated to improve application security and user management efficiency. Driver profile management functionalities were developed to store important information such as vehicle details, travel preferences, contact information, ride history, and seat availability. These profile management features allowed drivers to update and manage their ride-sharing information efficiently within the application. Proper database structures and backend APIs were implemented to support smooth handling of user records and profile management functionalities.

Ride Posting and Booking functionalities were developed to support the core operations of the Smart Ride Sharing System. Drivers could publish ride details, including source, destination, travel date, departure time, available seats, and preferences. Passengers were able to search rides based on route and travel requirements and make bookings through the platform. Booking confirmation features provided reservation details and ride status updates after successful transactions. This module established the foundation for efficient ride management and future system enhancements.

4.3.2 Milestone–2 : Fare Calculation, Payment & Route Matching Module

The second milestone focused on implementing dynamic fare calculation, secure online payment processing, and route matching functionalities within the Smart Ride Sharing System. Dynamic fare calculation was designed to automatically determine ride pricing based on travel distance, ride route, and passenger travel requirements. This functionality improved pricing transparency and ensured fair fare distribution between passengers and drivers. Fare calculation algorithms were integrated with ride booking functionalities to provide accurate ride cost estimation during the booking process. Secure online payment processing was implemented using Stripe card payment integration. The payment module enabled passengers to complete transactions securely using online card payment methods. Payment tracking features were also implemented to maintain transaction records, payment history, and booking-related financial information within the system. These functionalities improved reliability and transparency in payment management operations.

The project additionally included secure transaction handling mechanisms to protect payment information and ensure safe financial communication between users and the application. Backend APIs and database operations were integrated carefully to support real-time payment updates and maintain transaction consistency throughout the booking process. Stripe payment integration improved the overall usability and professionalism of the ride-sharing platform. Route matching and ride tracking functionalities were also developed during this milestone to improve transportation optimization and ride allocation efficiency. The system analyzed travel routes and matched passengers with drivers traveling in similar directions. Real-time ride updates and tracking-related information were implemented to improve coordination between passengers and drivers. These functionalities supported efficient shared mobility management, reduced unnecessary travel, and enhanced the overall ride-sharing experience within the platform.

4.3.3 Milestone–3 : Notification System, Review & Admin Dashboard

The third milestone focused on developing communication and user engagement features within the Smart Ride Sharing System. An in-app notification system was implemented to keep drivers and passengers informed about important ride-related activities. Users received notifications for booking confirmations, ride requests, payment updates, ride reminders, cancellations, and status changes. This functionality improved communication between users and reduced the chances of missing important updates. The notification module was integrated with multiple system components to ensure timely delivery of information. As a result, the overall user experience became more efficient and reliable. During this milestone, a review and rating system was also developed to encourage transparency and service quality within the platform. After successful ride completion, passengers and drivers were able to provide feedback based on their travel experience.

The review mechanism helped users evaluate service quality and build trust among platform members. Ratings and comments allowed both parties to maintain accountability and improve future interactions. This feature supported a more reliable and community-driven ride-sharing environment. It also contributed to enhancing user confidence while using the application. Another major achievement of this phase was the implementation of the Admin Dashboard for centralized monitoring and management. The dashboard enabled administrators to oversee users, rides, bookings, payments, notifications, and reviews through a single interface. Administrative functionalities were developed to monitor system activities, manage user records, and track operational performance efficiently. The dashboard also assisted in identifying issues, maintaining platform security, and ensuring smooth application operation. These monitoring capabilities improved overall system control and maintainability. This milestone significantly strengthened the management, communication, and quality assurance aspects of the Smart Ride Sharing System.

4.3.4 Milestone–4 : Testing, Review & Documentation

The final milestone of the Smart Ride Sharing System focused on testing, debugging, optimization, and documentation activities to ensure successful project completion. Different testing methods such as unit testing, integration testing, and user acceptance testing were performed to verify application functionality, security, usability, and overall system performance. Debugging and optimization tasks were carried out to resolve technical issues, improve response time, and maintain smooth execution of modules including authentication, ride booking, payment processing, notifications, and admin management functionalities. This milestone also included preparation of technical documentation, deployment guides, troubleshooting reports, and user manuals to support future maintenance and project understanding. Final system reviews and performance evaluations were conducted to ensure proper implementation of all project requirements and necessary improvements were applied wherever required. The completion of this phase improved application reliability, enhanced user experience, and provided practical exposure to real-world software testing, debugging, and professional project management practices during the internship.

Chapter 5

INTERNSHIP PROJECT DEVELOPMENT AND IMPLEMENTATION

The internship project development and implementation phase focused on designing, developing, integrating, and testing the Smart Ride Sharing System using modern full-stack technologies. The project was executed through a structured approach that included requirement analysis, system design, frontend and backend development, database integration, and security implementation. Various modules such as user management, ride booking, fare calculation, payment processing, notifications, and admin monitoring were successfully developed and integrated. Continuous testing and debugging activities were performed to ensure system reliability and performance. This phase provided valuable hands-on experience in real-world software development and project execution practices.

5.1 Development of the Smart Ride Sharing System

The implementation phase of the Smart Ride Sharing System focused on developing a secure, efficient, and user-friendly web application that connects drivers and passengers traveling in the same direction. The project implementation included frontend development, backend integration, database management, authentication handling, payment gateway integration, and system testing. The primary goal of this phase was to transform the project design into a fully functional ride-sharing platform capable of handling real-time user interactions and transportation-related operations. During the internship, modern software development methodologies and full-stack development practices were followed to ensure systematic implementation of all project modules. Different technologies such as React.js, Java Spring Boot, MySQL, JWT Authentication, and Stripe Payment Integration were combined to create a scalable and industry-oriented web application. The implementation process followed a modular structure where each module performed a specific functionality such as user management, ride booking, notifications, and payment processing. The implementation phase additionally improved practical understanding of real-world software engineering workflows including requirement analysis, coding, debugging, testing, deployment, and project documentation. Through continuous development and integration activities, the Smart Ride Sharing System evolved into a complete ride-sharing solution supporting secure communication, efficient ride management, and shared mobility services.

5.1.1 Overview of the Smart Ride Sharing System Implementation

The Smart Ride Sharing System was implemented as a full-stack web application developed to support efficient ride-sharing between drivers and passengers through a secure online platform. Drivers can publish ride details such as source location, destination, travel date, available seats, and ride preferences, while passengers can search and reserve rides according to their travel requirements. The implementation focused on creating a smooth, user-friendly, and reliable transportation management system that supports shared mobility and improves travel convenience. The frontend interface was developed using React.js along with HTML, CSS, and JavaScript to create responsive and interactive webpages for users. Backend functionalities were implemented using Java Spring Boot to manage business logic, authentication handling, ride booking operations, payment processing, route matching, and REST API communication. MySQL database services were additionally integrated to store and manage user records, ride details, booking information, notifications, reviews, and payment transactions securely throughout the application workflow. Additional functionalities such as JWT Authentication, Stripe card payment integration, email notification services, route-based ride matching, review management, and admin monitoring dashboards were also implemented during development. The project was designed to reduce travel costs, minimize traffic congestion, optimize vehicle usage, and promote environmentally sustainable transportation solutions while providing practical exposure to real-world full-stack application development methodologies.

5.1.2 Objectives of Project Development & Technologies Used During Implementation

The primary objective of the Smart Ride Sharing System was to develop a modern web-based transportation platform that connects drivers and passengers traveling in the same direction. The project aimed to reduce fuel consumption, transportation expenses, and unnecessary vehicle usage while supporting safe, affordable, and sustainable ride-sharing solutions. The implementation additionally focused on improving transportation coordination through functionalities such as ride posting, booking management, dynamic fare calculation, payment processing, notifications, and review management. Another important objective of the internship project was to gain practical exposure to full-stack web application development using modern technologies and software engineering practices. The project enhanced technical understanding of frontend development, backend integration, database communication, authentication handling, API development, debugging, testing, and deployment-related activities within a real-world development environment.

Multiple technologies and development tools were used during implementation to support efficient application development and management. React.js, HTML, CSS, and JavaScript were used for frontend development, while Java Spring Boot was implemented for backend services and REST API creation. MySQL was used for database management along with JPA Hibernate ORM for object-relational mapping. Additional technologies such as JWT Authentication, Maven, MVC Architecture, Stripe Payment Integration, and email notification services were also integrated to improve security, scalability, payment handling, and maintainability throughout the Smart Ride Sharing System.

5.2 Initial Product Understanding and System Analysis

The initial stage of the internship project focused on understanding the complete workflow of a ride-sharing platform and analyzing the requirements required for successful implementation of the Smart Ride Sharing System. Different transportation-related challenges such as ride coordination, fare management, passenger convenience, secure booking, and user communication were studied carefully before beginning development activities. System analysis activities were conducted to understand how frontend services, backend modules, database operations, and payment systems interact within a ride-sharing application. Existing transportation and mobility platforms were observed to identify common ride-sharing functionalities and improve project planning for the internship implementation process. This phase additionally helped identify user expectations and technical requirements for drivers, passengers, and administrators. Requirement gathering and workflow planning established the foundation for smooth development, proper module integration, and efficient implementation of all functionalities required within the Smart Ride Sharing System.

5.3 Requirement Analysis

Requirement analysis was performed to identify all functional and non-functional requirements required for successful implementation of the Smart Ride Sharing System. This phase focused on understanding application behavior, security needs, workflow operations, database communication, and frontend-backend integration before starting actual development activities. Different ride-sharing operations such as user registration, ride posting, booking management, payment processing, notifications, and review submission were analyzed carefully to define project modules and system architecture. Proper requirement analysis helped simplify implementation planning and improved project maintainability throughout the development lifecycle. The analysis process additionally ensured that the application satisfies both user expectations and technical requirements required for secure and scalable ride-sharing operations. Requirement documentation also helped maintain smooth communication between different modules during frontend, backend, and database integration activities.

5.3.1 Functional and Non-Functional Requirements

The Smart Ride Sharing System required multiple functional features to support smooth ride-sharing operations between drivers, passengers, and administrators. Important functionalities included secure user registration, login authentication, ride posting, ride searching, seat booking, booking confirmation, dynamic fare calculation, payment processing, route matching, notifications, and review management. These features were necessary for maintaining proper coordination and communication throughout the ride-sharing workflow. Different functionalities were designed according to user roles within the application. Drivers required ride publishing, seat availability management, booking tracking, and vehicle information handling functionalities, while passengers required ride searching, route-based filtering, seat reservation, payment processing, and review submission features. Administrative functionalities such as monitoring users, rides, bookings, payments, and reviews were additionally implemented through the admin dashboard to maintain centralized application management. The non-functional requirements focused on improving application security, reliability, scalability, responsiveness, and maintainability throughout the development process. JWT Authentication, password encryption, secure API communication, token expiration handling, and role-based access control were implemented to improve application security and protect sensitive user information. Responsive webpage design, efficient database operations, optimized backend performance, and reliable payment integration additionally improved system stability and user experience within the Smart Ride Sharing System.

5.3.2 User Requirements for Drivers, Passengers, and Admin

Drivers required functionalities that allowed them to register securely, manage personal and vehicle information, publish rides, update seat availability, and monitor passenger bookings through the Smart Ride Sharing System. The platform additionally needed to support real-time ride updates and proper coordination between drivers and passengers during ride-sharing operations. Passengers required functionalities such as secure account registration, ride searching, destination-based filtering, seat booking, payment processing, booking confirmation, notifications, and review submission after ride completion. These features were implemented to improve ride accessibility, booking convenience, and transportation management for passengers using the application. Administrators required centralized monitoring and management functionalities through the Admin Dashboard. The admin module supported management of users, rides, bookings, payments, reviews, and notifications within the platform. These user-specific requirements helped establish proper workflow implementation and improved overall management efficiency within the Smart Ride Sharing System.

5.4 System Design and Planning

The System Design and Planning phase focused on creating a structured architecture and organized workflow for the Smart Ride Sharing System to ensure smooth communication between frontend interfaces, backend services, database operations, authentication modules, and payment integration services. A layered and modular architecture was planned to improve application scalability, maintainability, security, and efficient ride-sharing operations. Frontend pages, backend APIs, authentication handling, notifications, payment processing, and dashboard functionalities were carefully designed before implementation to simplify project development and integration. The planning phase additionally included database schema creation, workflow organization, and module integration strategies required for managing ride posting, booking operations, route matching, fare calculation, payment tracking, notifications, and review management. MySQL database tables and relationships were designed to support secure data storage and efficient communication between different application modules. Proper workflow and module planning improved development efficiency, reduced implementation complexity, and supported successful execution of all functionalities within the Smart Ride Sharing System.

5.5 Frontend Implementation

The Frontend Implementation phase of the Smart Ride Sharing System focused on developing responsive and interactive web pages using React.js, HTML, CSS, and JavaScript. React.js was used to create reusable components such as dashboards, forms, ride cards, and navigation sections, while HTML, CSS, and JavaScript improved web page structure, styling, responsiveness, and user interaction. Integration with backend REST APIs enabled real-time data communication and provided a smooth user experience for drivers, passengers, and administrators across different devices and screen sizes.

5.5.1 User Interface Development Using React.js, HTML, CSS, and JavaScript

The frontend of the Smart Ride Sharing System was designed and developed using React.js, HTML, CSS, and JavaScript to deliver a modern, responsive, and easy-to-use web application. React.js enabled the creation of reusable interface components such as authentication forms, navigation panels, dashboards, ride listings, and booking modules, which improved code reusability and simplified frontend maintenance. The interface was carefully structured to provide convenient access to system features for drivers, passengers, and administrators. Emphasis was placed on creating an intuitive design that enhanced usability, accessibility, and overall user satisfaction. Consistent layouts, organized content placement, and responsive design techniques contributed to a professional appearance across the application. These frontend technologies played a significant role in delivering a smooth and efficient user experience. The Home Page served as the main gateway to the platform and introduced users to the key services offered by the Smart Ride Sharing System. It contained important sections such as ride-sharing information, service highlights, navigation options, and direct access to registration, login, ride search, and ride posting features. A dedicated Chatbot was also incorporated to provide instant assistance and guide users through various platform functionalities. The chatbot supported users by answering frequently asked questions, explaining booking procedures, and helping them navigate different system features. This functionality improved user engagement and provided quick support without requiring manual intervention. Together, the Home Page and Chatbot enhanced accessibility, convenience, and interaction within the platform as shown in Fig. 5.1.

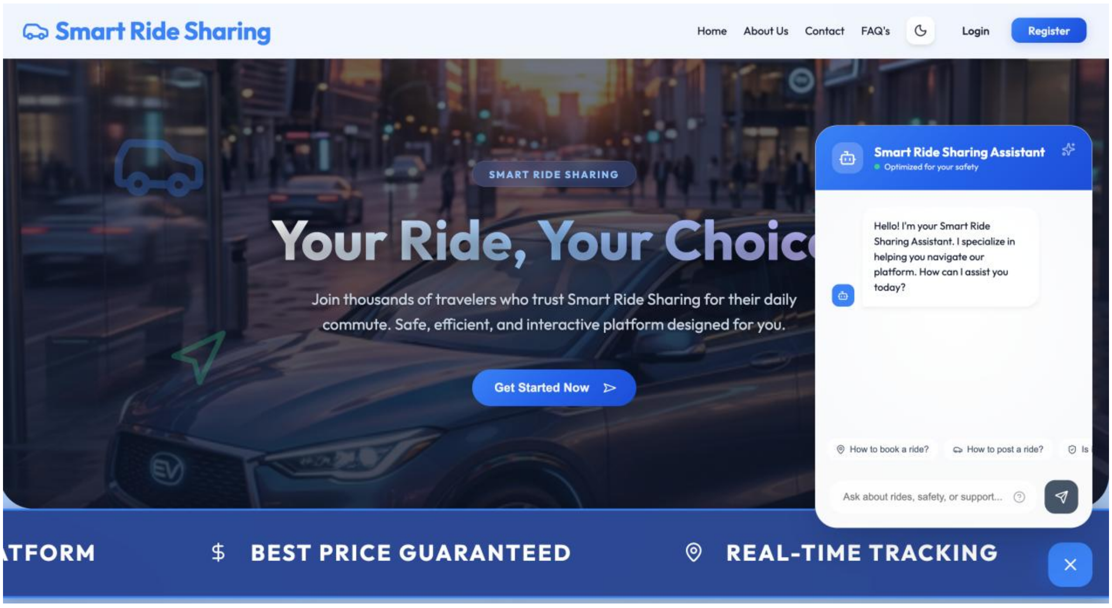


Fig. 5.1: Home Page and Chatbot

HTML was utilized to define the structure of web pages, while CSS was applied to create visually appealing layouts and ensure compatibility across different screen sizes. JavaScript was used to implement interactive features such as form validation, dynamic content updates, notifications, and user-driven actions. Through React.js, frontend components communicated efficiently with backend REST APIs to retrieve and display ride details, booking information, notifications, and user records in real time. Dynamic rendering techniques improved responsiveness by minimizing page reloads and ensuring seamless interaction. The integration of these technologies established effective communication between frontend and backend modules. As a result, the Smart Ride Sharing System delivered a reliable, interactive, and user-focused web interface.

5.5.2 Responsive Web Pages Design Implementation

The responsive web pages of the Smart Ride Sharing System were developed to provide a seamless and user-friendly experience across different devices and screen sizes. Technologies such as React.js, HTML, CSS, and JavaScript were utilized to build dynamic and interactive interfaces. The design focused on improving accessibility, usability, and navigation throughout the platform. Responsive layouts ensured that users could access system functionalities efficiently from desktops, tablets, and mobile devices. This approach enhanced overall user satisfaction and application performance. The Home Page was designed as the primary entry point of the application, offering users a clear overview of the platform and its services. It included navigation menus, service highlights, ride-sharing information, and quick access links to important features.

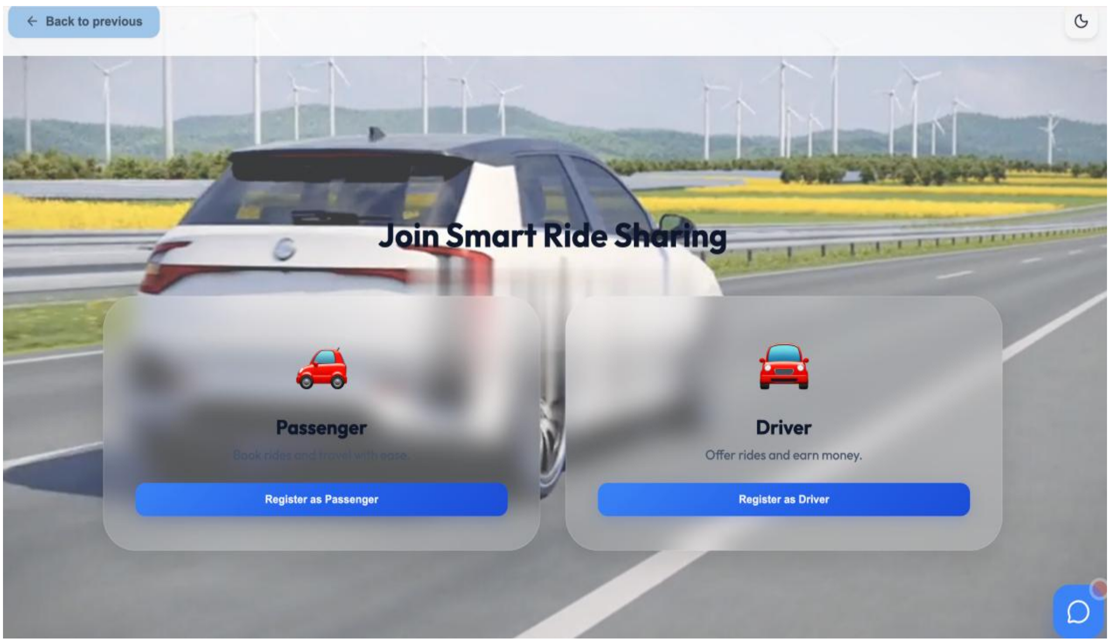


Fig. 5.2: Registration Page

Interactive sections were incorporated to improve user engagement and simplify navigation. The page was structured to help users easily understand the purpose and functionality of the platform. This design contributed to a more organized and intuitive user experience.

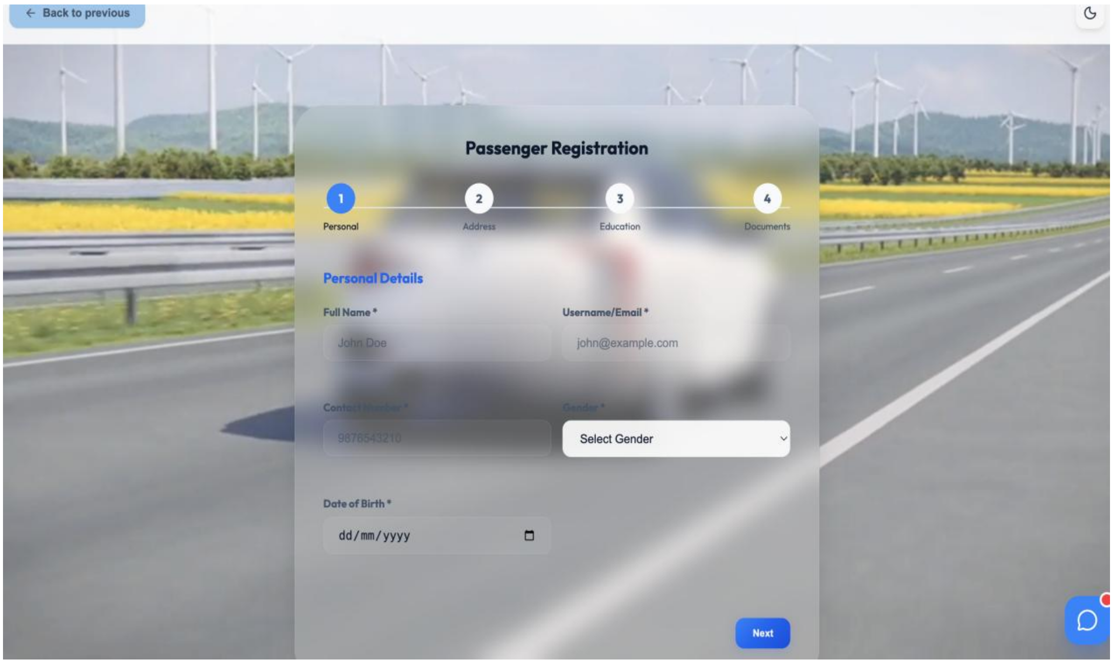


Fig. 5.3: Passenger Registration Page

Role-based registration functionality was implemented to support separate account creation processes for drivers and passengers. Users were required to provide personal information, residential address details, educational qualifications, and identity verification documents during registration as shown in Fig. 5.3. Documents such as Aadhaar Card or PAN Card could be uploaded for account verification purposes. This process improved user authenticity and strengthened platform security. Secure validation mechanisms were also implemented to ensure accurate data submission. The Login and Registration Pages were designed with responsive layouts and user-friendly forms to simplify account access and management. Input validation techniques were incorporated to reduce errors and improve data accuracy during authentication. The interface was optimized to provide a smooth registration and login experience across different devices as shown in Fig. 5.4. Clear instructions and structured form designs enhanced usability for new and existing users. These features contributed to a secure and convenient authentication process.

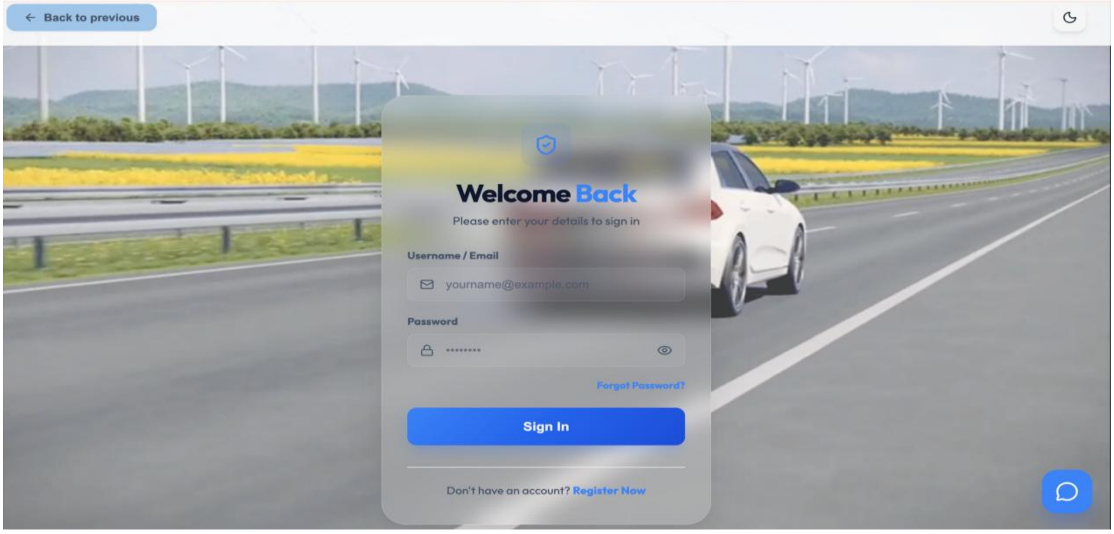


Fig. 5.4: Login Page

The Contact Page of the Smart Ride Sharing System was developed to provide users with a convenient way to communicate with the platform support team and report any issues or queries. As shown in Fig. 5.5, the page contains a structured contact form where users can enter details such as Full Name, Email Address, Phone Number, Query Type, Subject, and Message Description. This information helps the support team understand user concerns and respond effectively. An attachment upload feature was also integrated into the form, allowing users to submit screenshots or supporting documents related to their issues.

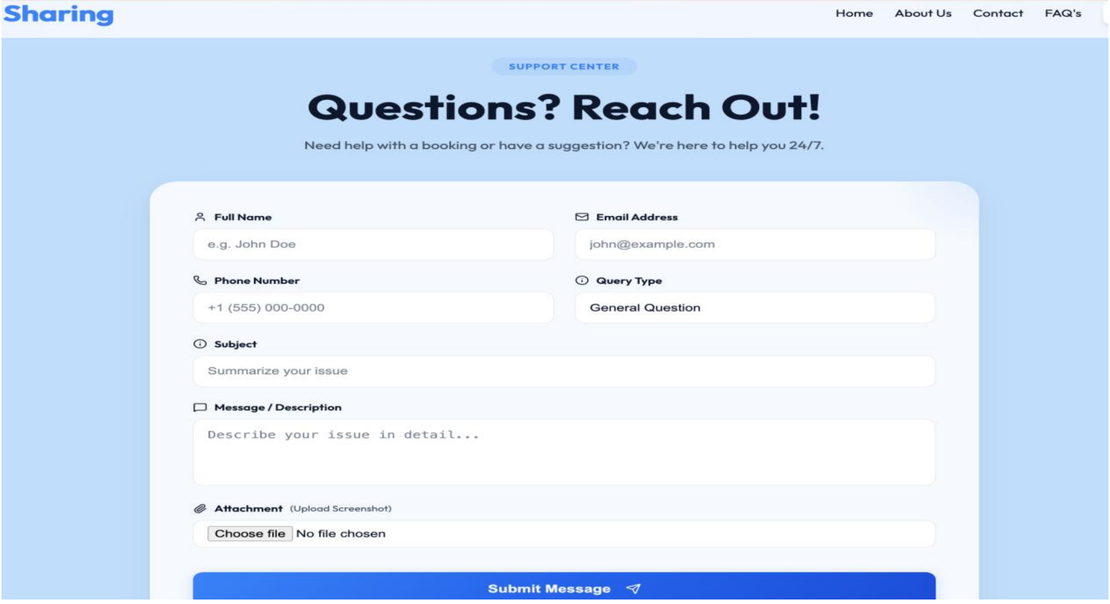


Fig. 5.5: Contact Page

This functionality improves problem identification and helps administrators provide faster and more accurate assistance. Different query categories were included to organize requests efficiently and simplify the support process. The Contact Page was designed with a clean, responsive, and user-friendly interface to ensure accessibility across desktops, tablets, and mobile devices. Form validation mechanisms were implemented to verify mandatory fields and prevent incomplete submissions. All submitted queries are securely stored in the database and can be reviewed through the administrative dashboard for proper tracking and resolution. This page enhances communication, improves customer support services, and contributes to a better user experience within the Smart Ride Sharing System.

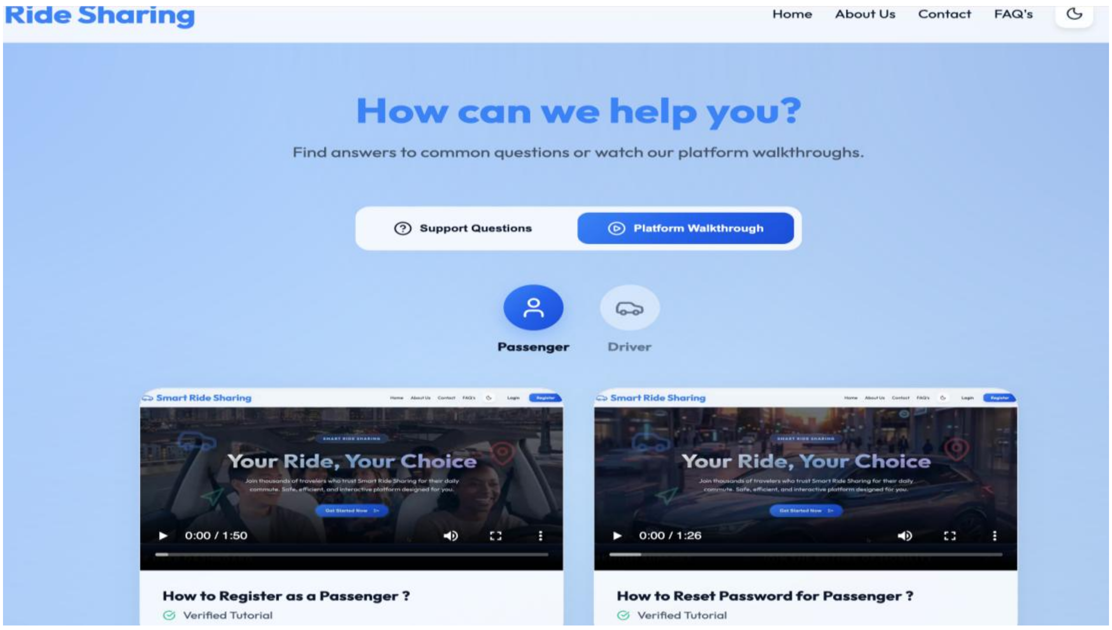


Fig. 5.6: FAQ's Page

The FAQ's Page, Contact Page and Chatbot interface were developed to provide guidance and support for both drivers and passengers. The FAQ section included commonly asked questions related to ride booking, ride posting, payments, account management, and platform usage. Walkthrough videos were also provided to help users understand various system functionalities more effectively as shown in Fig. 5.6. The Chatbot offered quick responses to common queries and assisted users in navigating the platform. Together, these support features improved user assistance, reduced confusion, and enhanced the overall experience of using the Smart Ride Sharing System.

5.5.3 Dashboard Design Implementation

Separate dashboards were implemented for Admin, Driver, and Passenger functionalities within the Smart Ride Sharing System. The Driver Dashboard supports ride posting, booking management, seat availability updates, and ride history tracking functionalities throughout the application workflow as shown in Fig. 5.7.

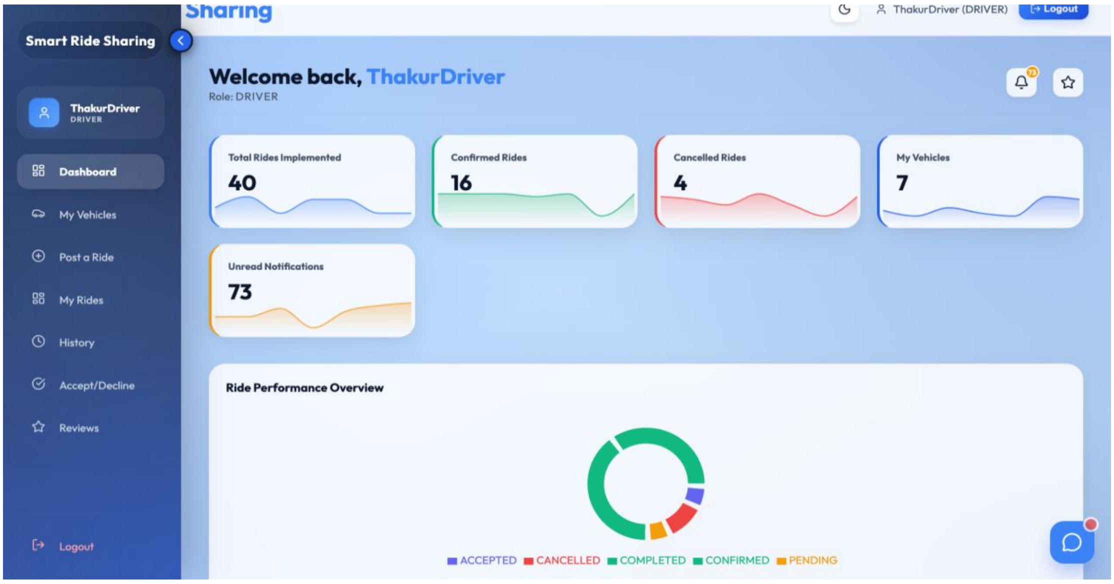


Fig. 5.7: Driver Dashboard

The Passenger Dashboard was developed to support ride searching, booking management, payment tracking, notifications, and review submission functionalities.

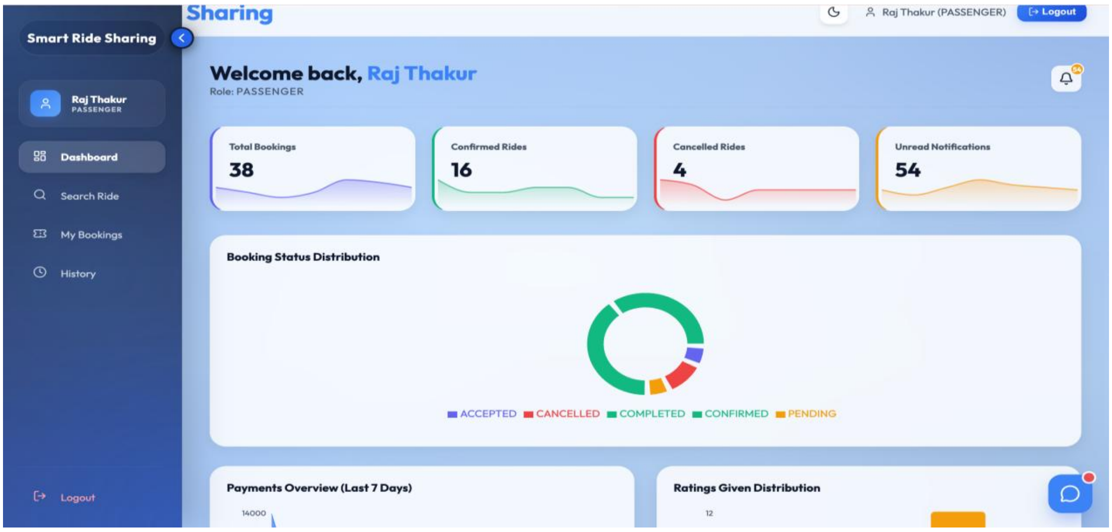


Fig. 5.8: Passenger Dashboard

Interactive dashboard components and real-time updates improved user convenience and simplified ride-sharing operations for passengers using the platform as shown in Fig. 5.8. The Admin Dashboard was implemented to provide centralized monitoring functionalities for users, rides, payments, bookings, reviews, and overall application activities. Dashboard implementation improved administrative control, ride management efficiency, and application monitoring throughout the system lifecycle as shown in Fig. 5.9.

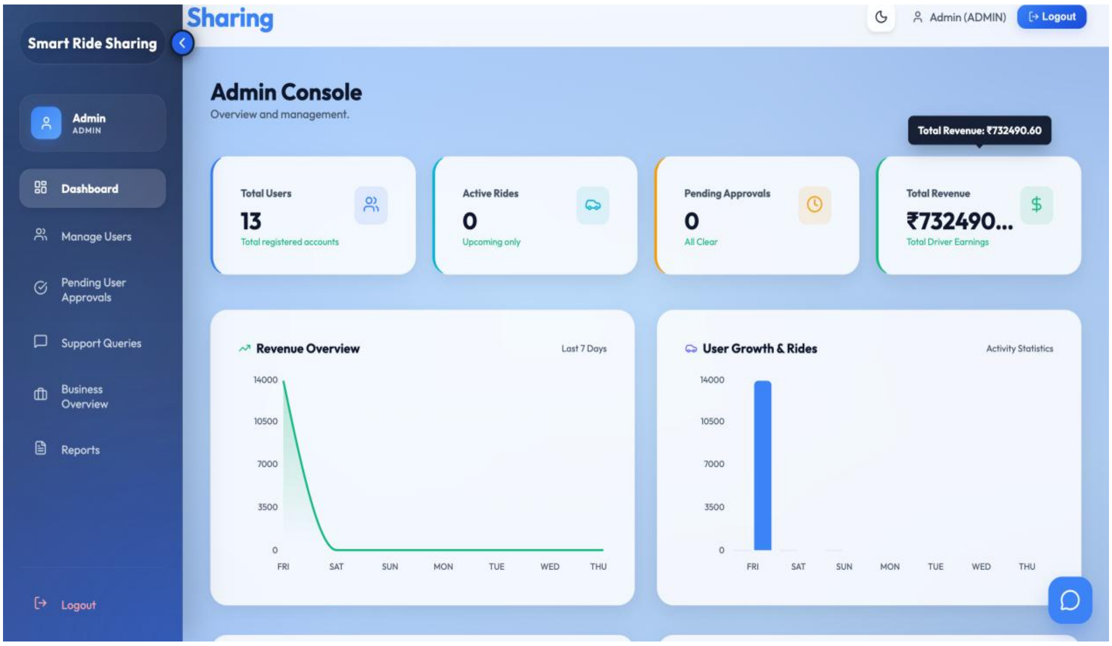


Fig. 5.9: Admin Dashboard

5.5.4 Email Integration Implementation

Email integration was implemented in the Smart Ride Sharing System to provide reliable communication between passengers, drivers, and administrators. Automated email services were configured to send important notifications related to account management, ride booking activities, and support requests. These email functionalities improved user awareness by delivering timely updates directly to registered email addresses. Backend email services were integrated using secure SMTP configurations to ensure dependable message delivery. This communication mechanism enhanced the overall efficiency and professionalism of the platform. A Passenger Account Approval Email feature was developed to notify passengers once their registration details and verification documents were reviewed and approved by the administrator.

This email informed users about successful account activation and provided confirmation that they could access platform services as shown in Fig. 5.10.

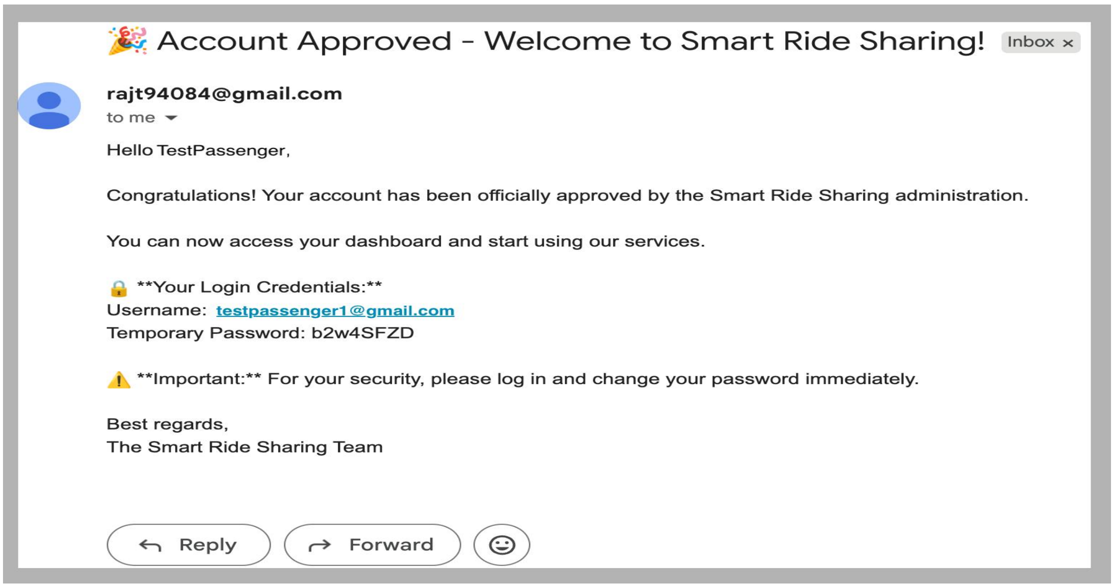


Fig. 5.10: Passenger Account Approval Email

The approval process improved account security and ensured that only verified users could participate in ride-sharing activities. Automated notifications reduced manual communication efforts and improved user onboarding. This functionality streamlined the registration workflow within the system.

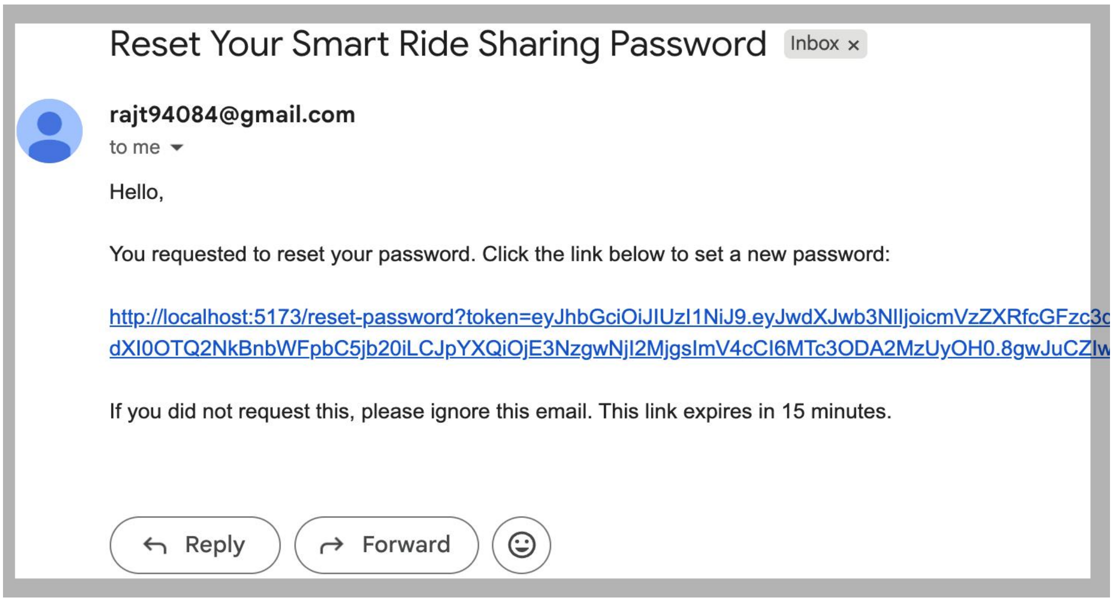


Fig. 5.11: Forgot Password Reset Email

A Forgot Password Reset Email service was implemented to assist users in recovering access to their accounts securely. When a user requested a password reset, the system generated a unique reset token and sent a secure password recovery link to the registered email address. This process allowed users to create a new password while maintaining account security as shown in Fig. 5.11.

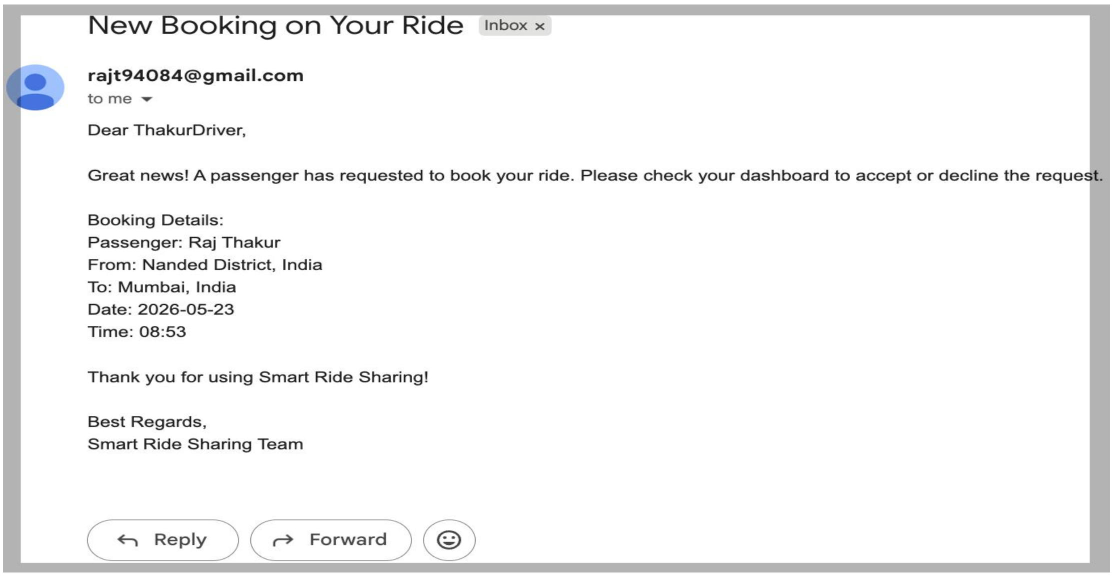


Fig. 5.12: Ride Booking Request Email

Token validation mechanisms were used to prevent unauthorized access during password recovery. The feature improved user convenience and reduced login-related support requests.

Fig. 5.13: Ride Booking Confirmation Email

Ride Booking Request Emails and Ride Booking Confirmation Emails were integrated to keep both drivers and passengers informed throughout the booking process. Drivers received notifications whenever a passenger submitted a booking request, while passengers received confirmation emails after their bookings were approved. These automated messages included essential ride-related information to improve coordination and transparency between users. Real-time communication reduced uncertainty and improved booking management. The email workflow contributed to a smoother ride-sharing experience for all participants as shown in Fig. 5.12 and Fig. 5.13.

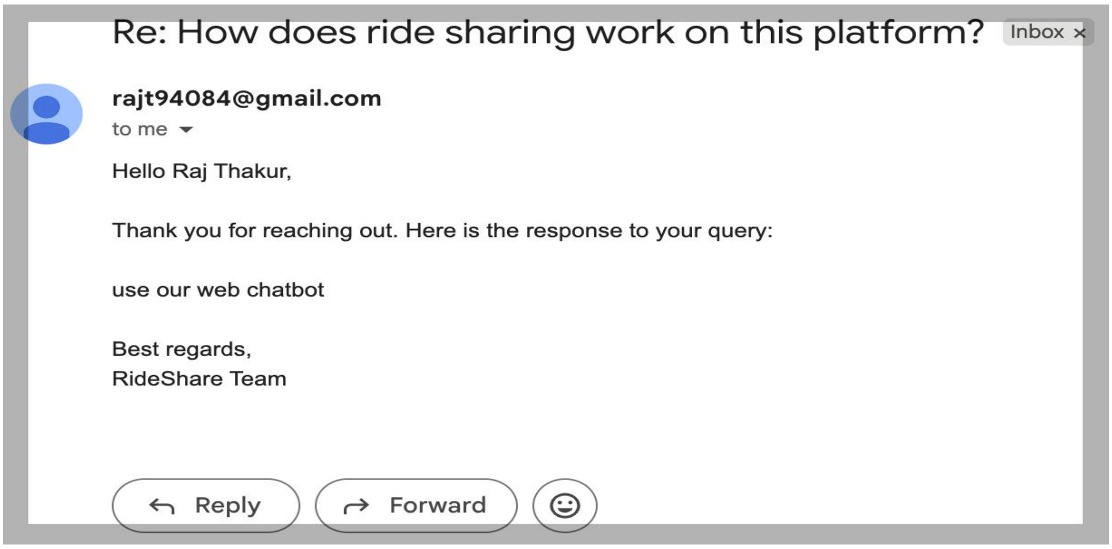


Fig. 5.14: Contact & Support Email

The Contact and Support Email feature was implemented to manage user queries, feedback, and support requests submitted through the Contact Page. Automated email notifications ensured timely communication between users and administrators, improving issue tracking, response efficiency, and overall customer support within the Smart Ride Sharing System as shown in Fig. 5.14.

5.6 Backend Implementation

The Backend Implementation phase focused on developing server-side functionalities and application logic using Java Spring Boot within the Smart Ride Sharing System. Backend services handled important operations such as authentication, ride posting, booking management, payment processing, notifications, and database communication. Spring Boot was selected because of its scalability, security support, and efficient enterprise-level development capabilities. Backend services were organized into controllers, services, and repositories to improve maintainability and simplify communication between frontend components and database operations. REST APIs were additionally implemented to establish secure communication between frontend interfaces and backend services. Proper backend implementation improved application reliability, security, and smooth management of ride-sharing functionalities throughout the internship project.

5.6.1 Java Spring Boot Backend Development & REST API Integration

Java Spring Boot was implemented as the primary backend framework for developing the server-side functionalities of the Smart Ride Sharing System. The framework simplified backend configuration, dependency management, and application development while supporting scalable and secure full-stack implementation throughout the project. Backend modules were developed to manage functionalities such as user authentication, ride posting, booking processing, fare calculation, notifications, payment integration, and review management using separate Controller, Service, and Repository layers for better code organization and maintainability. REST APIs were also created to establish smooth communication between React.js frontend interfaces and backend services, where API endpoints handled operations such as registration, login, ride searching, booking confirmation, and payment processing through HTTP methods including GET, POST, PUT, and DELETE. This backend implementation improved frontend-backend synchronization, application scalability, and reliable execution of ride-sharing functionalities across the platform.

5.6.2 MVC Architecture Implementation & Business Logic Handling

The Smart Ride Sharing System followed the MVC (Model-View-Controller) Architecture to maintain proper separation between frontend, backend, and database functionalities while improving application organization and scalability. The Model layer managed database entities such as users, rides, bookings, payments, and reviews, the Controller layer handled frontend requests and REST API communication, and the View layer represented interactive frontend pages and user interfaces. Business logic was implemented within the Service layer to process ride posting, booking validation, route matching, fare calculation, payment tracking, notification management, and review handling operations. Additional functionalities such as JWT token validation, seat availability verification, ride status management, and secure transaction processing further improved application reliability, maintainability, security, and smooth ride-sharing operations throughout the system.

5.7 Database Implementation

The Database Implementation phase focused on managing application data securely and efficiently within the Smart Ride Sharing System. MySQL was used as the relational database management system for storing user records, ride details, booking information, payments, notifications, and reviews throughout the application workflow. Database operations were integrated with backend services using JPA Hibernate ORM to simplify communication between Java entities and database tables. Structured database implementation improved data consistency, reduced redundancy, and supported efficient management of ride-sharing operations within the platform. The database layer additionally improved secure data handling and reliable retrieval operations required for authentication, booking management, payment tracking, and administrative monitoring functionalities throughout the system lifecycle.

5.7.1 Users Table

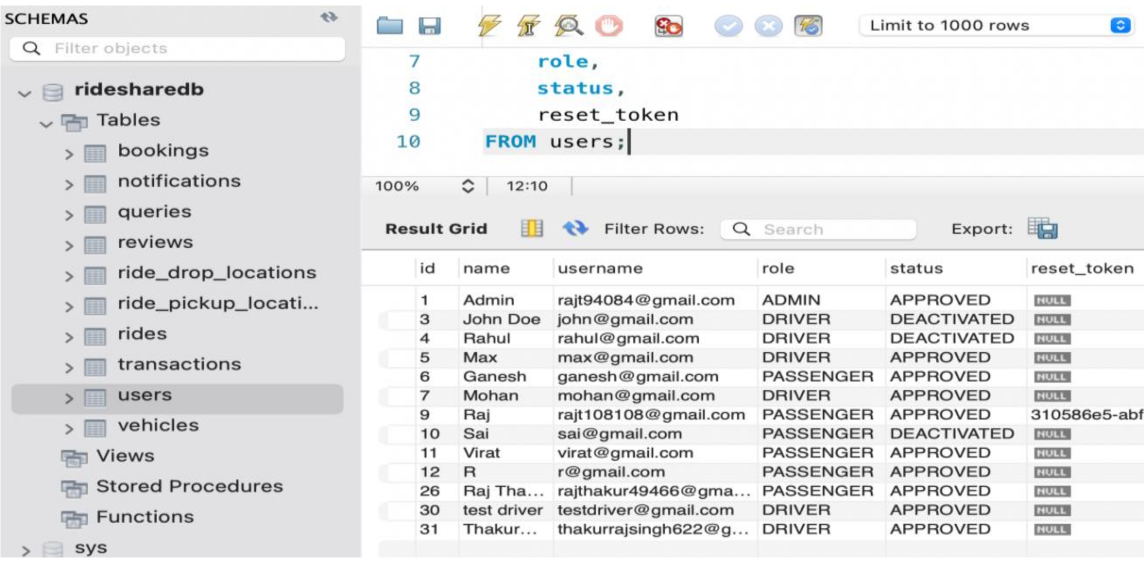


Fig. 5.15: Users Table

The **users** table was implemented to manage complete user information for drivers, passengers, and administrators within the Smart Ride Sharing System. As shown in Fig. 5.15. It stored authentication details such as USERNAME, PASSWORD, ROLE, STATUS, RESET_TOKEN, and IS_FIRST_LOGIN along with personal details including NAME, DATE_OF_BIRTH, CONTACT_NUMBE4R, AADHAR_CARD_ID, GENDER, and PAN_CARD_ID. Educational details such as SCHOOL_10, SCHOOL_12, TENTH_PERCENTAGE, TWELFTH_PERCENTAGE, GRADUATION_COLLEGE, and GRADUATION_YEAR were additionally maintained for profile verification and record management purposes.

5.7.2 Vehicles Table

The **vehicles** table was created to maintain vehicle-related information associated with registered drivers in the Smart Ride Sharing System. This table stored details such as VEHICLE_NUMBER, COMPANY, MODEL, COLOR, CAPACITY, MANUFACTURING_YEAR, KM_DRIVEN, and vehicle feature information including HAS_AC and HAS_AUDIO_SYSTEM. Insurance and registration verification details such as INSURANCE_DETAILS, INSURANCE_IMAGE_URL, RC_NUMBER, and RC_IMAGE_URL were also managed within this table for secure vehicle validation.

5.7.3 Transactions Table

The **transactions** table was designed to manage secure payment and financial transaction records generated during ride bookings. Important fields such as AMOUNT, BOOKING_ID, PAYMENT_METHOD, STATUS, and TRANSACTION_TIME were used to maintain payment tracking and transaction history within the platform. This table supported Stripe card payment integration and ensured reliable financial management throughout the ride-sharing workflow.

5.7.4 Rides Table

The **rides** table stored all ride posting information published by drivers within the application. Fields including SOURCE, DESTINATION, DATE_TIME, AVAILABLE_SEATS, DISTANCE, PRICE, STATUS, DRIVER_ID, and

VEHICLE_ID were implemented to manage travel routes, ride schedules, fare details, and seat availability efficiently. This table played an important role in ride searching, booking management, and transportation coordination throughout the Smart Ride Sharing System.

5.7.5 Ride Pickup Locations Table

The **ride_pickup_locations** table was implemented to store pickup point information related to different rides within the application. Fields such as ride_id, LOCATION, and location_order were used to organize multiple pickup locations according to ride routes and travel flow. This table improved route management and allowed passengers to select suitable pickup points during ride booking operations.

5.7.6 Ride Drop Locations Table

The **ride_drop_locations** table was developed to manage destination and drop point information for passengers within the Smart Ride Sharing System. Fields including ride_id, LOCATION, and location_order helped organize drop locations in proper sequence according to travel routes. This table supported accurate ride tracking and improved route-based transportation coordination throughout the platform.

5.7.7 Reviews Table

The **reviews** table was created to manage passenger and driver feedback after ride completion. Important fields such as RATING, COMMENT, CREATED_AT, REVIEWER_ID, REVIEWEE_ID, and RIDE_ID were implemented to store ratings and review information associated with completed rides. This table improved transparency, service quality monitoring, and user trust within the Smart Ride Sharing System.

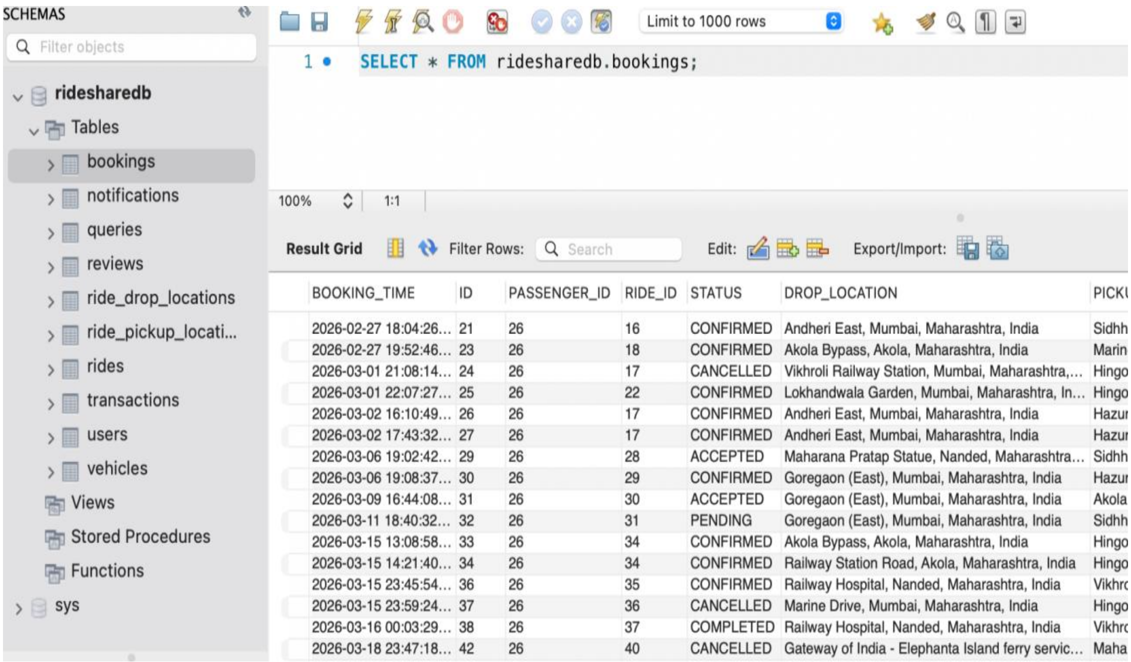
5.7.8 Queries Table

The **queries** table was implemented to manage user support requests, contact messages, and issue reporting functionalities within the application. Fields such as EMAIL, MESSAGE, SUBJECT, QUERY_TYPE, STATUS, ADMIN_RESPONSE, ATTACHEMENT_URL, FULL_NAME, PHONE_NUMBER, and RIDE_ID were used to maintain communication records between users and administrators. This table supported customer support and issue management services throughout the platform.

5.7.9 Notifications Table

The **notifications** table was designed to manage notification and alert services within the Smart Ride Sharing System. Fields including USER_ID, MESSAGE, IS_READ, and CREATED_AT were implemented to store booking confirmations, ride reminders, payment updates, and ride-related alerts for users. This table improved communication flow and supported real-time system updates throughout the application.

5.7.10 Bookings Table



BOOKING_TIME	ID	PASSENGER_ID	RIDE_ID	STATUS	DROP_LOCATION	PICKU
2026-02-27 18:04:26...	21	26	16	CONFIRMED	Andheri East, Mumbai, Maharashtra, India	Sidhh
2026-02-27 19:52:46...	23	26	18	CONFIRMED	Akola Bypass, Akola, Maharashtra, India	Marin
2026-03-01 21:08:14...	24	26	17	CANCELLED	Vikhroli Railway Station, Mumbai, Maharashtra,...	Hingo
2026-03-01 22:07:27...	25	26	22	CONFIRMED	Lokhandwala Garden, Mumbai, Maharashtra, In...	Hingo
2026-03-02 16:10:49...	26	26	17	CONFIRMED	Andheri East, Mumbai, Maharashtra, India	Hazur
2026-03-02 17:43:32...	27	26	17	CONFIRMED	Andheri East, Mumbai, Maharashtra, India	Hazur
2026-03-06 19:02:42...	29	26	28	ACCEPTED	Maharana Pratap Statue, Nanded, Maharashtra...	Sidhh
2026-03-06 19:08:37...	30	26	29	CONFIRMED	Goregaon (East), Mumbai, Maharashtra, India	Hazur
2026-03-09 16:44:08...	31	26	30	ACCEPTED	Goregaon (East), Mumbai, Maharashtra, India	Akola
2026-03-11 18:40:32...	32	26	31	PENDING	Goregaon (East), Mumbai, Maharashtra, India	Sidhh
2026-03-15 13:08:58...	33	26	34	CONFIRMED	Akola Bypass, Akola, Maharashtra, India	Hingo
2026-03-15 14:21:40...	34	26	34	CONFIRMED	Railway Station Road, Akola, Maharashtra, India	Hingo
2026-03-15 23:45:54...	36	26	35	CONFIRMED	Railway Hospital, Nanded, Maharashtra, India	Vikhrc
2026-03-15 23:59:24...	37	26	36	CANCELLED	Marine Drive, Mumbai, Maharashtra, India	Hingo
2026-03-16 00:03:29...	38	26	37	COMPLETED	Railway Hospital, Nanded, Maharashtra, India	Vikhrc
2026-03-18 23:47:18...	42	26	40	CANCELLED	Gateway of India - Elephanta Island ferry servic...	Maha

Fig. 5.16: Bookings Table

As shown in Fig. 5.16. The **bookings** table was developed to store ride reservation and booking-related information within the Smart Ride Sharing System. Important fields such as PASSENGER_ID, RIDE_ID, BOOKING_TIME, STATUS, PICKUP_LOCATION, DROP_LOCATION, and SEATS were implemented to manage passenger bookings and seat reservations efficiently. This table supported booking confirmation, ride tracking, and transportation coordination throughout the ride-sharing workflow.

5.8 Ride Posting & Booking Implementation

The Ride Posting and Booking Implementation phase focused on developing functionalities that allow drivers to publish rides and passengers to search and reserve available seats through the Smart Ride Sharing System. This module was designed to improve transportation coordination by connecting users traveling in the same direction through a secure and user-friendly web platform. Proper integration between frontend interfaces, backend services, and database modules ensured smooth ride management and booking operations throughout the application. The implementation additionally supported real-time ride availability updates, booking confirmations, and seat management functionalities for both drivers and passengers. Secure communication between frontend React.js components and backend REST APIs improved application responsiveness and enabled efficient ride-sharing operations throughout the platform workflow.

5.8.1 Driver Ride Posting Functionality

The Driver Ride Posting Functionality was implemented to allow drivers to publish travel information within the Smart Ride Sharing System. Drivers could enter details such as source location, destination, travel date, available seats, vehicle information, pickup points, drop locations, and ride pricing while creating ride postings through the application dashboard.

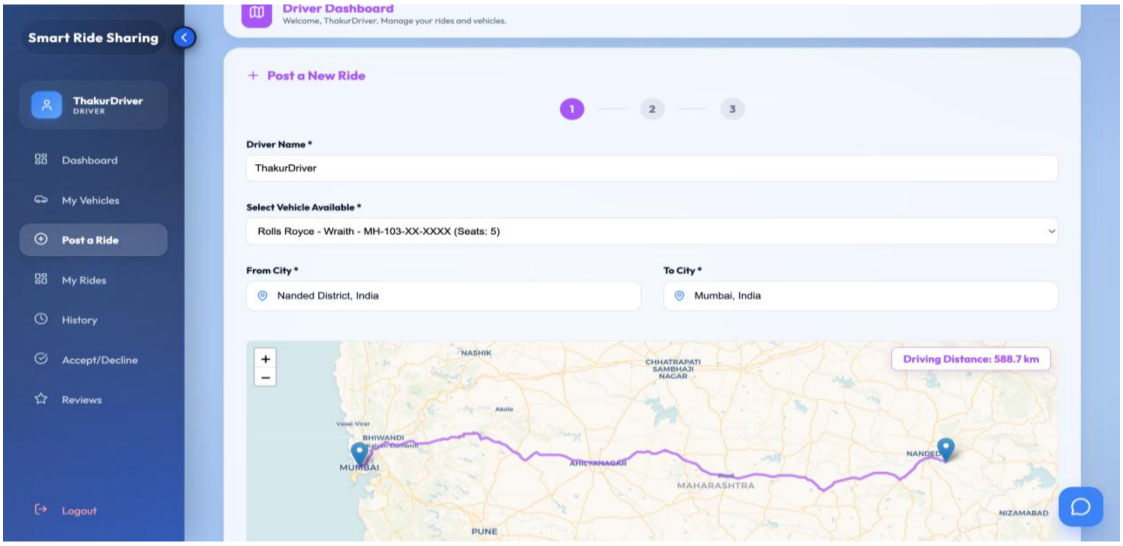


Fig. 5.17: Driver Ride Posting Functionality

Backend APIs and database services were integrated to store ride information securely and manage ride availability dynamically throughout the platform. This functionality improved ride coordination and enabled drivers to efficiently share transportation resources with passengers traveling along similar routes as shown in Fig. 5.17.

5.8.2 Passenger Ride Search and Booking Functionality

The Passenger Ride Search and Booking Functionality was implemented to help passengers search for available rides according to their travel destinations and booking preferences. Passengers could filter rides based on source, destination, travel date, and seat availability using responsive search interfaces developed within the application. After selecting a suitable ride, passengers could reserve seats and receive booking confirmation details through secure backend processing and database integration. This functionality improved booking convenience, simplified ride accessibility, and enabled efficient communication between passengers and drivers throughout the ride-sharing workflow.

5.9 Fare Calculation, Payment, Notification & Review System

The Fare Calculation, Payment, Notification, and Review System Implementation phase focused on improving financial management, communication, and user feedback functionalities within the Smart Ride Sharing System. These modules were integrated together to provide transparent ride pricing, secure payment handling, real-time notifications, and service quality monitoring throughout the application workflow. The implementation additionally improved user interaction and ride coordination by enabling automated fare estimation, booking alerts, payment updates, and review submission functionalities. Integration of payment gateways, notification services, and review management systems enhanced the reliability and usability of the ride-sharing platform.

5.9.1 Fare Calculation and Payment Integration

The Fare Calculation and Payment Integration module was developed to provide automatic ride fare estimation and secure online payment processing within the application. Dynamic fare calculation was implemented based on travel distance and ride routes to ensure transparent and fair pricing for passengers and drivers. Stripe card payment integration was additionally implemented to support secure online transactions and maintain payment records within the system.

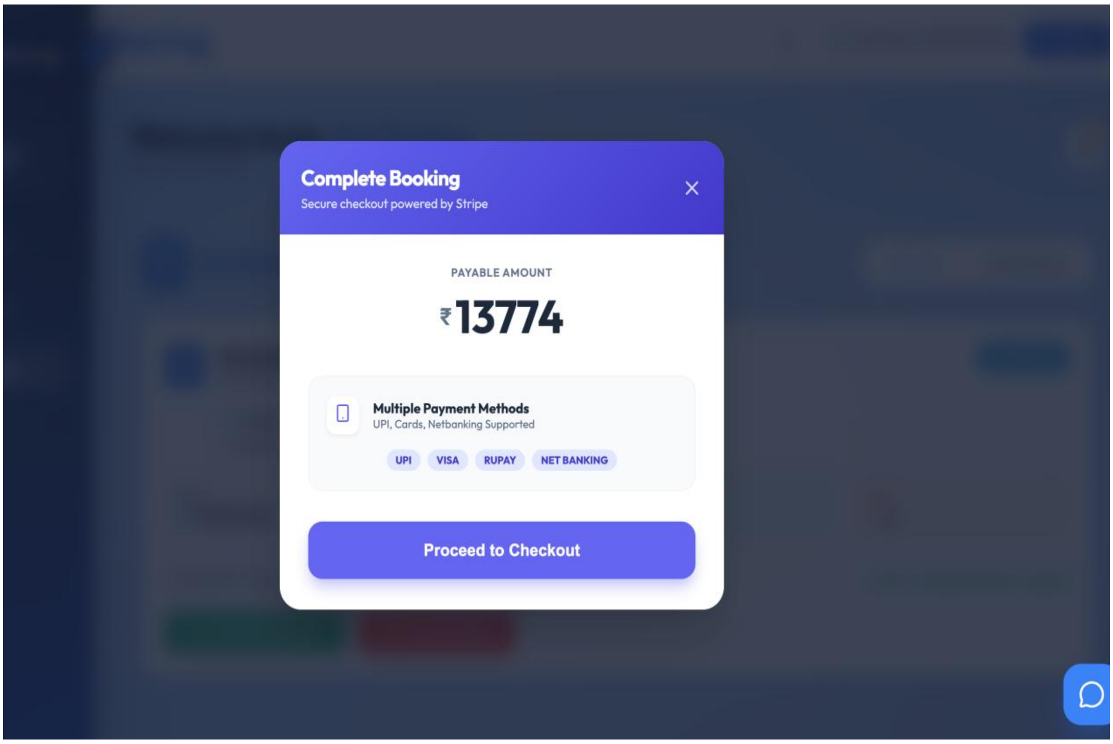


Fig. 5.18: Payment through Stripe

The module also managed payment tracking, booking confirmation after successful transactions, and transaction history management to improve financial reliability and user trust throughout the Smart Ride Sharing System as shown in Fig. 5.18.

5.9.2 Notification System & Review and Rating Management

The Notification System was implemented to provide real-time updates related to ride bookings, payment confirmations, ride reminders, cancellations, and booking status changes within the Smart Ride Sharing System. Notification services improved communication between passengers, drivers, and administrators while maintaining smooth coordination throughout ride-sharing operations. The Review and Rating Management module allowed passengers and drivers to submit feedback and ratings after successful ride completion. This functionality improved transparency, service quality monitoring, and trust between users while helping maintain better ride-sharing experiences within the platform.

5.10 Testing & Debugging

The Testing and Debugging phase focused on verifying the stability, functionality, and security of the Smart Ride Sharing System before final deployment. Different testing methods were performed to ensure that modules such as user authentication, ride booking, payment processing, notifications, and review management worked correctly without affecting application performance. Frontend and backend functionalities were tested together to validate smooth communication between React.js interfaces, Spring Boot APIs, and MySQL database operations. Debugging activities were additionally carried out to identify and resolve issues related to API responses, database connectivity, authentication handling, payment integration, and frontend responsiveness. Proper error handling mechanisms and validation techniques were implemented to improve system reliability, application security, and overall user experience throughout the ride-sharing workflow.

5.11 Challenges Faced During Internship

During the internship project, several technical and implementation-related challenges were encountered while developing the Smart Ride Sharing System. Integrating frontend interfaces with backend APIs, managing secure JWT authentication, handling role-based access control, and maintaining real-time ride updates required careful planning and debugging throughout the development process. Payment gateway integration and database relationship management also presented additional complexities during implementation. Different problem-solving approaches were used to overcome these challenges, including modular development practices, API testing, debugging techniques, database optimization, and proper validation handling. Continuous testing, code restructuring, and implementation reviews helped improve application performance, security, and maintainability throughout the internship project development lifecycle. Ensuring seamless communication between multiple system modules while maintaining data consistency and application reliability was another significant challenge addressed during the project development process.

5.12 Outcomes of Project Implementation

The implementation of the Smart Ride Sharing System provided valuable practical exposure to modern full-stack web application development methodologies and software engineering practices. The project improved understanding of frontend development using React.js, backend implementation using Java Spring Boot, REST API integration, MySQL database management, JWT authentication, and secure payment processing within a real-world application environment. The internship project additionally enhanced technical confidence, debugging abilities, teamwork coordination, and problem-solving skills related to enterprise-level software development. Successful completion of the project improved knowledge of scalable application architecture, secure system implementation, and professional development workflows used in modern transportation and ride-sharing platforms.

5.12.1 Skills Gained During Implementation

During the implementation phase, multiple technical and professional skills were developed through practical involvement in the Smart Ride Sharing System project. Full-stack development skills related to React.js, Java Spring Boot, REST APIs, MySQL, JPA Hibernate ORM, JWT Authentication, and Stripe payment integration were significantly improved throughout the internship experience. The project additionally enhanced debugging techniques, API testing, database management, frontend-backend integration, and responsive web application design skills. Exposure to real-world software development practices also improved logical thinking, code organization, problem-solving ability, and understanding of modular application architecture.

5.12.2 Real-World Software Development Experience

The internship project provided real-world software development experience by involving complete application development activities from requirement analysis to testing and deployment planning. Practical exposure to frontend implementation, backend services, database integration, authentication handling, payment gateway integration, and application debugging improved understanding of professional development workflows used in industry-level projects. Working on the Smart Ride

Sharing System also improved knowledge of collaborative development practices, project planning, secure coding standards, and system scalability concepts. This real-time project experience helped strengthen practical implementation abilities and prepared for future opportunities in full-stack web application development and software engineering domains. The project enhanced problem-solving abilities by requiring the analysis and resolution of various technical challenges during development. Hands-on experience with modern development tools and frameworks improved overall technical proficiency and coding efficiency. Implementing different system modules provided a deeper understanding of software architecture and component interaction. Regular testing and debugging activities strengthened skills in identifying, troubleshooting, and resolving application issues. The internship also improved documentation, time management, and project execution skills essential for professional software development environments.

CONCLUSION

During this internship, I gained practical experience in full-stack web application development by working on the project titled “Development of a Dynamic Ride-Sharing and Carpooling Platform” called as Smart Ride Sharing System using technologies such as React.js, Java Spring Boot, MySQL, REST APIs, HTML, CSS, JavaScript, JWT Authentication, and JPA Hibernate ORM. I improved my understanding of frontend design, backend development, database management, authentication handling, and API integration while implementing real-time ride-sharing functionalities within the application.

I also gained valuable knowledge in project planning, debugging, testing, payment integration, notification management, and responsive web page development. Working on different modules of the system helped me improve my technical skills, logical thinking, teamwork, and problem-solving abilities while understanding real-world software engineering and development practices followed in industry projects.

Overall, the internship enhanced my confidence in developing secure and scalable web applications and strengthened my practical knowledge of modern software development technologies. The experience provided hands-on exposure to real-world project implementation and helped me build a strong foundation in full-stack development and professional application design.

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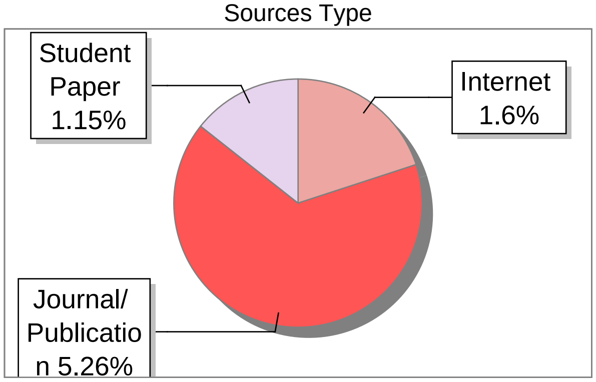

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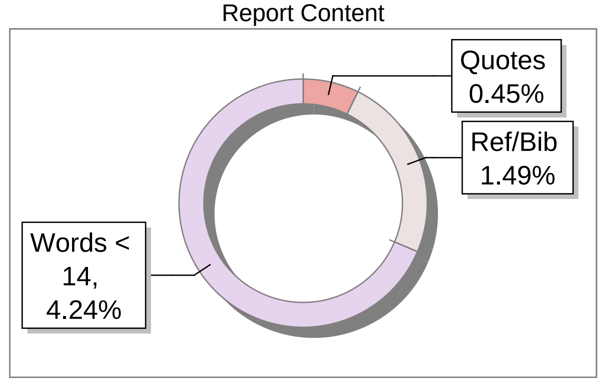

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